

All-Possibility Hierarchical Filtering Simulation (APHFS): A Resource-Bounded Framework from Executable Rules toward Multiscale Biology, Aging, and Rejuvenation Research

Mingyuan Chen

Department of Biochemistry and Molecular Biology, Johns Hopkins Bloomberg School of Public Health,

Johns Hopkins University, Baltimore, Maryland, USA

Correspondence: a515006420@gmail.com

3 September 2026

Abstract

Scientific models begin as imagined explanations; mathematics makes them executable, whereas evidence determines which remain tenable. All-Possibility Hierarchical Filtering Simulation (APHFS) joins bottom-up candidate generation to top-down reality constraints. “All” is local: within a declared finite grammar, resource envelope, and precision policy, every address receives a terminal account. The framework preserves observational ambiguity, rejects an inadequate class only under complete simultaneous bounds, and otherwise permits an indeterminate result. It withholds coarse-graining until specified upper-level observables and decisions remain stable under refinement. Future fault-tolerant quantum computers could enlarge executable dynamics, but would be an execution substrate rather than evidence.

We evaluated this machinery once in the complete 256-rule elementary-cellular-automaton grammar. The generating rule and operational signature class were retained in all 64 A1 blocks, although five blocks retained two rules and three retained two classes. A deliberately distinguishable radius-two construction returned model-class inadequacy in all 64 B1 blocks. D2-CERT used the provably density-preserving periodic Rule 170 shift: 0/64 calibration and 0/64 held-out violations checked certificate-pipeline conformance for a constructed closed map, not uncertain coarse-law accuracy. Four resource policies reached identical decisions. One prespecified salted pseudo-random early-stopping order was least costly on this constructed workload (75,080 units), versus 89,816 for the other order, 92,592 for adaptive fidelity, and 132,096 for exhaustive search.

These results establish bounded, failure-capable behavior only for this synthetic finite grammar, not physical or biological validation. A separately prespecified enzyme-kinetics study is the next empirical bridge toward multiscale biology; aging and safety-constrained rejuvenation remain longer-term destinations.

1 Introduction

Model discovery faces a problem that extends beyond search accuracy. A procedure may find the best model it was allowed to consider while the whole language is wrong; sparse observations may support several mechanisms; and a numerical approximation may preserve low-level values yet alter the upper-level conclusion. A credible search method must expose all three possibilities.

Imagine a person ten thousand years ago who sees the round Sun and proposes that Earth might also be round, only to be mocked by companions. The round Sun does not prove a spherical Earth, and ridicule does not refute it. The episode serves one purpose here: to separate hypothesis generation from evidential validation. An imagined

rule must first be expressed precisely enough to generate consequences and then confronted with observations independent of its formulation.

Core thesis. APHFS connects imagination, executable mathematics, and evidence through a resource-bounded sequence of decisions. Bottom-up generation explores a declared language; top-down observations constrain each bridge toward higher-level descriptions. Within a fixed finite grammar, resource envelope, and precision policy, every executable candidate is enumerated or explicitly accounted for. Observationally equivalent candidates remain ambiguous, a valid simultaneous test may find the entire class inadequate, and missing support produces an indeterminate result rather than a convenient conclusion. Coarse-graining is not accepted before its domain, memory, and error conditions are checked. Lower-level fidelity is sufficient only when prespecified refinements preserve registered upper-level observables and decisions. Computation can expand what is evaluated; evidence constrains what can be justified. A future fault-tolerant quantum execution layer could make some quantum-native candidate dynamics tractable, while adding explicit state-preparation, error-correction, measurement, and classical-post-processing obligations. It would not relax complete accounting or turn computation into empirical support. Biology, aging, and rejuvenation remain downstream empirical destinations.

The motivating intuition is conditional rather than ontological: if progressively expanded candidate languages are sufficiently expressive for the registered observables, generation and search reach relevant candidates, independent evidence is sufficiently discriminating, and estimation, numerical, dynamical, and coarse-graining errors are controlled, then the retained set may approach the lowest observable risk attainable within the declared comparison envelope. This does not guarantee recovery of nature’s ontology. A grammar may omit the relevant mechanism, finite observations may retain a set, an entire class may be inadequate, and each new domain requires new evidence.

The phrase *All-Possibility* is therefore local. It means complete accounting within a stated grammar, encoding, bit budget, resource envelope, and precision policy—not enumeration of every law or ontology in nature. Enlarging the language creates a new claim. This boundary separates a finite scientific object that can be audited from an unrestricted metaphysical assertion.

The framework also separates questions that leaderboards merge. Candidate ranking asks which member fits best. Operational signatures ask which members the observation interface can distinguish. A class-level gate asks whether even the best-supported class is inadequate. If a candidate, bound, or error-budget entry is missing, the class decision is **INDETERMINATE**; absence is never imputed as evidence against the class.

The long-term destination is biological. The proposed route starts with executable rules below fixed particle labels and advances only through empirical bridges to physical, chemical, molecular, and cellular organization. Particle-like candidates must face physical and scattering constraints; atom-like candidates must face atomic spectra; and molecule-like candidates must face molecular structure, energy, and reaction data. Whole-cell studies illustrate both the reach of integrated mechanistic models and the difficulty of validating emergent predictions [Karr et al., 2012, Thornburg et al., 2022, 2026a,b]. Aging would enter when a model can predict long-horizon homeostatic drift and declining resilience under appropriate longitudinal and perturbational data [López-Otín et al., 2013, 2023]. Rejuvenation research would require causal interventions with efficacy, persistence, toxicity, off-target effects, uncontrolled proliferation, and cancer risk made explicit. These are downstream research goals, not interpretations of cellular automata.

Here, *pre-particle* is a modeling stance, not an empirical claim of a newly discovered subparticle ontology. Candidate rules may precede fixed particle labels in the generative description, but particle-like status must be earned through stable observables and independently specified physical constraints. Quantum cellular automata are an important future candidate family, not a conclusion forced by the current ECA study.

Beyond the finite ECA demonstration, candidate languages may vary dimension, geometry, locality, symmetry, interaction rules, motion, stochasticity, composition, and response to perturbation. When a declared finite grammar is affordable, APHFS requires genuine enumeration and complete terminal accounting. When it is not, stratified, randomized, sampled, or adaptive generation may be used only with its coverage frame and missing mass stated; such a search is not relabeled as local-All enumeration.

This paper integrates six capabilities needed for such model search: advance definition of the candidate language,

complete accounting, ambiguity-preserving signatures, a whole-class inadequacy gate, observable-level error and fidelity controls, and a one-time held-out evaluation. Systems biology provides the motivating decision structure because it must compare mechanistic explanations under partial observation, misspecification, coarse-graining, and finite compute. The current experiment uses all 256 elementary cellular automata (ECA) as an inspectable testbed in which every candidate can receive a terminal account. It evaluates the machinery, not biology.

2 The APHFS research program

Figure 1 places the current experiment inside the larger research program. Candidate construction proceeds bottom-up, beginning with executable rules rather than assuming the higher-level objects it ultimately seeks to explain. Constraints enter top-down as empirical anchors. Each transition—to persistent structures, particle-like signatures, molecular systems, cellular dynamics, aging, or intervention—is a new scientific question that can fail.

Two controls govern movement up the ladder. First, a coarse description is accepted only when its domain, memory, and error conditions support the proposed upper-level inference. Second, numerical resolution is decision-relative. A fidelity that preserves water density may not preserve a vibrational spectrum; one that preserves a ball’s center-of-mass trajectory may not preserve surface turbulence. More microscopic precision is useful only when it can change a registered observable, classification, decision, or certificate.

Aging and rejuvenation therefore occupy the far end of the path. Aging could be represented by long-horizon distributions of recovery, viable-state return, functional persistence, and entry into dangerous attractors. Rejuvenation would become a safety-constrained control problem over combination, order, timing, dose, persistence, and adverse effects, drawing on robust-control precedents [Iyengar, 2005, Nilim and El Ghaoui, 2005]. Neither interface is implemented here. Progress depends on new domain models and observations, not on computational scale alone.

From imagination through executable mathematics to evidence

Compute expands evaluation; evidence constrains justification.

Future FTQC expands executable dynamics; it does not supply evidence.

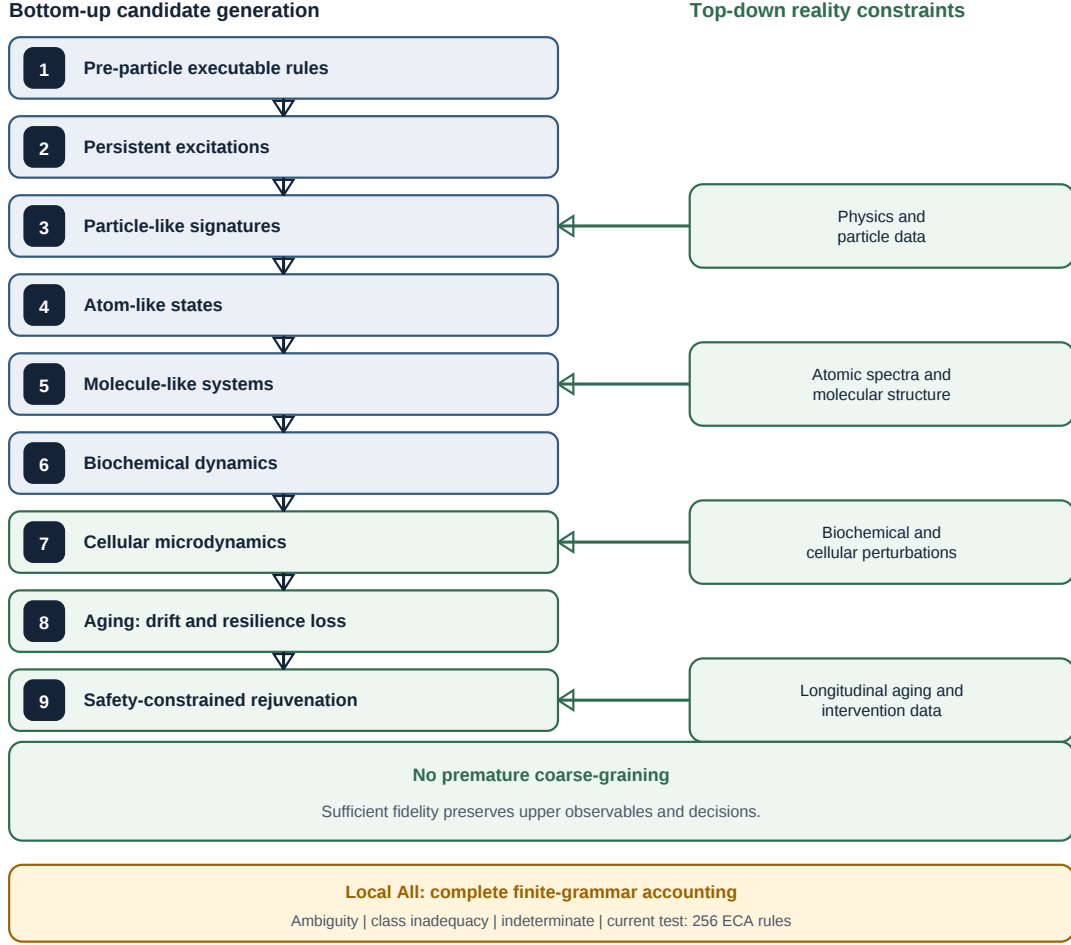


Figure 1: The APHFS research program. Imagination is encoded as executable candidates, complete finite-grammar accounting makes the search auditable, and empirical constraints determine which claims survive. Operational ambiguity, model-class inadequacy, and indeterminacy remain distinct. Higher-level objects require validated coarse-graining and protected upper-observable fidelity. Future fault-tolerant quantum hardware could expand executable candidate dynamics but would not supply evidence. The present evidence concerns only the 256-rule ECA machinery test; biology, aging, and rejuvenation remain downstream destinations.

3 How APHFS works

The six decisions in Figure 2 form a dependency chain: each supplies a premise for the next, and each has a distinct failure meaning.

3.1 Step 1: Define scope

The address alphabet, finite address set, single-valued parser, execution semantics, invalid-state rules, resources, precision policy, probes, and failure actions are fixed before evaluation. This turns an open-ended idea into a finite scientific object and reveals its expressive limits. Ledger identity is the declared address (or a predeclared canonical address), so syntactic aliases remain separate rows unless canonicalization was fixed in advance. Post-outcome changes create a new study; they cannot expand the old completeness claim.

APHFS: six scientific decisions

Each output becomes a premise for the next step; failure narrows the claim.

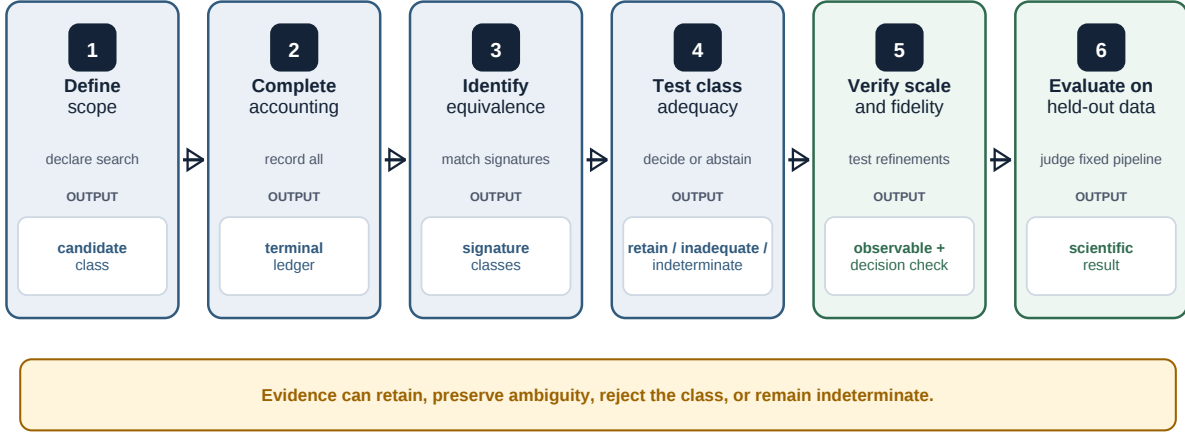


Figure 2: Six linked decisions in APHFS. Scope precedes complete accounting; operational equivalence precedes class-adequacy testing; scale and fidelity checks protect the inference; and held-out evaluation judges the fixed pipeline. Failure narrows the claim, returns an indeterminate result, or motivates a newly specified study.

The declaration therefore answers two questions before any ranking begins: which mechanisms are expressible, and which executions are affordable. Invalid syntax, invalid semantics, and exhausted resources remain visible outcomes. If a relevant mechanism cannot be represented, the limitation belongs to the candidate class; retained models do not become exhaustive over nature.

3.2 Step 2: Complete accounting

Every addressable candidate must be executed or receive an explicit terminal record, including invalid syntax, numerical failure, and resource exhaustion. Here all 256 ECA rule IDs were accounted for. A0 compared rule tables and trajectories under periodic, fixed-zero, fixed-one, and edge-replicate boundaries (historical code token: **reflect**); A1 used its separate block-level interface. The edge-replicate convention sets each out-of-range neighbor to the nearest boundary cell, rather than reflecting the next interior cell. A0’s scalar reference and NumPy-vectorized production paths were separately implemented within the same AI-assisted project and do not share their rule-lookup or one-step update functions. Their agreement is an internal implementation-conformance check, not a separately authored implementation or external replication. An absent candidate breaks whole-class inference. A partial run may still be useful, but its sampling frame must be stated.

This accounting is the premise for every later statement about the declared class. An unrecorded timeout or silently imputed failure breaks that premise because an unseen candidate could change the conclusion. Computational scarcity may narrow the searchable scope; it cannot assign a favorable value to missing evidence.

3.3 Step 3: Identify operational equivalence

Fixed probes and deterministic quantizers map each candidate to a canonical signature. Exact serialized equality partitions candidates that the observation interface cannot distinguish; candidate losses remain separate unless equality holds across the full evaluation interface. The result is set-valued identification, not ontological identity. A changed probe, tolerance, or serialization rule defines a new partition.

Operational equivalence makes ambiguity measurable. More informative observations may split a signature class, but an arbitrary representative cannot make the unresolved alternatives disappear. Equality under finitely many

probes therefore supports claims about that interface alone, not agreement on unseen inputs or a unique underlying mechanism.

3.4 Step 4: Test class adequacy

Ranking always produces a least-bad member, so APHFS asks whether simultaneous lower loss bounds place every candidate above a prespecified threshold. Bounds are formed at candidate level and aggregated conservatively through signature classes. Whole-class inadequacy is reported only with complete candidate or class coverage and a complete alpha ledger. A missing run, invalid bound, unsafe deduplication, or incomplete ledger returns `INDETERMINATE`, never evidence against the class.

In the present B0/B1 workload every class was a singleton and every loss was computed on complete finite support. Thus the class minimum and the general repeated-look allocation were definitions of the wider framework, not empirically exercised features of these endpoints. The observed B outcomes test exact-loss decision and completeness machinery for their constructed cases.

A valid inadequacy result has a different consequence from a poor winner: it sends the search back to the language or resource envelope before further ranking is useful. Conversely, incomplete uncertainty control requires abstention. The procedure is designed to distinguish a bad model from a bad language and both from insufficient evidence.

3.5 Step 5: Verify scale and fidelity

Trajectory bounds track propagated local error; cross-scale bounds compare evolve-then-coarse-grain with coarse-grain-then-evolve on declared domains; memory checks test whether a coarse state supports prediction. Each fidelity record names the lower-level refinement, protected upper-level observable, tolerance, horizon, input or environment distribution, decision stability, and need for further refinement. Domain exit, missing output, or an observable or decision breach blocks a sufficiency claim. Raw datatype agreement is not enough.

The scientific question is not whether two floating-point arrays are numerically close in the abstract, but whether refinement can change the registered conclusion. Microscopic discrepancies may contract or amplify, whereas further precision may be unnecessary when every protected quantity and decision remains stable. Sufficiency is always relative to the stated interface.

3.6 Step 6: Evaluate on held-out data

Development constructs the language, probes, thresholds, policies, and diagnostics. Calibration served only the prespecified D2-CERT event. A role-separated held-out evaluation then judged the fixed pipeline once, without retuning or retry. This separation is internal to the project; it is not external replication. Leakage, outcome-dependent threshold changes, result replacement, or repeated execution would void that interpretation. Stable computation is a premise for evidence, not evidence by itself.

This separation keeps evaluation data from becoming another optimization signal. Calibration can support only the distribution-conditional statement for which it was designed; the held-out step judges the completed pipeline. Its first outcome remains the outcome whether favorable, unfavorable, or indeterminate, so failure cannot be erased by rerunning the decision.

The complete formal configuration can be written as

$$\mathfrak{A} = (\mathcal{G}, \text{parse}, \text{sem}, \text{invalid}, \text{enc}, \rho, b, r, \Pi, \mathcal{P}, Q, \mathcal{S}, \mathcal{E}),$$

where the entries encode grammar and execution rules, description and resource limits, precision, probes, the calibration law, split manifest, and global error budget. Its purpose is not to make the workflow look formal; it makes every evidential premise inspectable and prevents silent substitution of a different scientific object.

4 Mathematical foundations

The equations below formalize the points at which the narrative could otherwise become vague. Table 1 gives a reader’s map: what each expression is for, how to read it, and which failure removes its conclusion. Detailed derivations and domain conditions appear in the Supplement.

Table 1: Reader’s map to the core equations.

Equation	Scientific question	How to read it	What withdraws it
(1)	How many possibilities are actually in scope?	A finite address set and single-valued parser bound the executable semantic image; the ledger still accounts for every declared address.	Parser/address-policy changes or incomplete accounting.
(2)	Why may the selected model remain worse than the best declared comparator?	Approximation, sampling, regularization, and optimization contribute separately.	Adapted class or loss, dependence, unbounded loss, or a comparator outside the declared premise.
(3)	What can the fixed observations distinguish?	Exact serialized equality creates operational equivalence classes, not ontological identity.	Changed probes or quantizers, pairwise tolerance, or unsafe representative-only evaluation.
(4)	When may the entire declared language be rejected?	Even the weakest valid lower bound in every class must exceed the threshold; current B blocks exercise only the singleton/exact-loss case.	Any missing candidate or bound, unsafe deduplication, or incomplete alpha ledger.
(5)	How can local error affect a later state?	Each discrepancy is multiplied by the sensitivity of the dynamics that follow it.	Domain exit, mismatched paths or metrics, or a missing local bound.
(6)	Do fine and coarse descriptions evolve consistently?	Local commutation defects accumulate through later coarse-map sensitivities.	Type or domain failure, or a missing Lipschitz premise.
(7)	When can calibration grant a distribution-conditional certificate?	Under the iid reference law, grant only if the exact one-sided upper endpoint is at most the target; the implemented conditioned law has the stated coverage correction.	Event or distribution change, pseudoreplication, role reuse, or tuning.
(8)	When does role-separated held-out evidence withdraw that certificate?	Withdraw only if the exact held-out lower endpoint exceeds the target.	A broken certificate chain, role reuse or broken role separation, or a changed rule.
(9)	How are repeated looks charged to one error budget?	Candidate, gate, and look receive prespecified shares, and the radius uses cumulative sample size n_s ; current endpoints did not call this machinery.	Unplanned looks or added candidates, dependence without a valid replacement, or an incomplete ledger.
(10)	When is the current precision enough for the scientific decision?	Every prespecified refinement must preserve both upper-level observables and decisions.	A tolerance or decision breach, missing output, or unfinished refinement.

4.1 Turning “all” into a finite statement

Let b be the maximum binary address length, let $\mathcal{A}_b = \bigcup_{j=0}^b \{0,1\}^j$, and fix before evaluation a single-valued parser $\text{parse} : \mathcal{A}_b \rightarrow \mathcal{H} \cup \{\text{INVALID}\}$. For resource envelope r , the executable semantic image is

$$\mathcal{H}_{b,r} = \{\text{parse}(a) : a \in \mathcal{A}_b, \text{parse}(a) \neq \text{INVALID}, \rho(\text{parse}(a)) \preceq r\}.$$

The ledger nevertheless ranges over addresses a : invalid and resource-infeasible addresses receive terminal rows, and semantic duplicates remain separate rows unless a canonical-address policy was declared before evaluation. The parser-image premise gives

$$|\mathcal{H}_{b,r}| \leq |\mathcal{A}_b| = \sum_{j=0}^b 2^j = 2^{b+1} - 1. \quad (1)$$

The equation should be read as a scope guarantee, not a promise of cheap computation. At $b = 100$, the address count is approximately 2.54×10^{30} : finite, auditable in definition, and still far too large for naive enumeration. Invalid or out-of-resource descriptions may reduce the executable set, but their addresses still require explicit ledger entries; semantic duplicates may reduce the image or the number of operational classes without reducing the declared addresses that must be accounted for. Completeness is therefore relative to this parser/address policy. If the language changes after evaluation or a subset is described as exhaustive, the counting claim no longer supports the scientific conclusion.

Counting also clarifies the gap between a selected model and the best comparator inside the declared envelope. For a common measurable loss $\ell(h, Z) \in [0, 1]$, Hoeffding’s bounded-loss concentration inequality supplies the finite-class sampling term used below [Hoeffding, 1963]. Define

$$R(h) = \mathbb{E}[\ell(h, Z)], \quad \hat{R}_n(h) = \frac{1}{n} \sum_{i=1}^n \ell(h, Z_i), \quad \varepsilon_{\text{app}}(b, r) = R_{b,r}^* - R_\infty^*.$$

Here $R_{b,r}^*$ is the minimum risk in $\mathcal{H}_{b,r}$, while R_∞^* is the infimum in a nonempty comparison envelope containing that class.

Theorem 1 (Finite-class APHFS specialization). *Let $n \geq 1$, $0 < \delta_{\text{conf}} < 1$, and $b \in \mathbb{N}_0$. Suppose observations are iid from one fixed distribution. The nonempty finite class $\mathcal{H}_{b,r}$, nonempty comparison envelope $\mathcal{H}_\infty \supseteq \mathcal{H}_{b,r}$, and one common measurable $[0, 1]$ -valued loss interface are fixed before and without using the evaluation sample. Assume $0 \leq L_d(h) \leq b$, $\lambda_n \geq 0$, and let $\hat{h}_n$ satisfy the explicit approximate-optimization condition*

$$\hat{R}_n(\hat{h}_n) + \lambda_n L_d(\hat{h}_n) \leq \min_{h \in \mathcal{H}_{b,r}} \{\hat{R}_n(h) + \lambda_n L_d(h)\} + \xi_n, \quad \xi_n \geq 0.$$

The following bound then holds with probability at least $1 - \delta_{\text{conf}}$:

$$R(\hat{h}_n) - R_\infty^* \leq \varepsilon_{\text{app}}(b, r) + 2\sqrt{\frac{(b+2)\log 2 + \log(1/\delta_{\text{conf}})}{2n}} + \lambda_n b + \xi_n. \quad (2)$$

The four terms tell four different stories. $\varepsilon_{\text{app}}(b, r)$ is what the declared language and resources cannot approximate; the square-root term is finite-sample uncertainty; $\lambda_n b$ is the cost of the chosen complexity penalty; and ξ_n is unfinished optimization. A future, prospectively specified increase in language or resources may reduce the first term, more complete search may reduce the fourth, and more iid evaluation observations may reduce the second; none turns the comparator into “nature.” Adapting the class, loss, distribution, or budget on the evaluation sample makes the theorem inapplicable. The ECA study checked machinery in one finite grammar; it did not empirically validate this standard finite-class specialization. Along a prespecified sequence, if the approximation, sampling, penalty, and optimization terms vanish and $\delta_{\text{conf},n} \rightarrow 0$, the excess risk converges in probability to zero relative to the declared comparison envelope. Almost-sure convergence additionally requires a summable tail schedule, such as $\sum_n \delta_{\text{conf},n} < \infty$. Neither conclusion covers grammar expansion adapted to the same evaluation sample.

4.2 Representing what observations cannot distinguish

Let $\mathcal{P} = (p_1, \dots, p_J)$ be fixed probes, $S_h(p_j)$ a typed output summary, and q_j a deterministic quantizer with fixed precision, boundary, tie-breaking, missing-value, and serialization rules. The canonical signature is

$$\begin{aligned} \sigma_{\mathcal{P}}(h) &= (q_1(S_h(p_1)), \dots, q_J(S_h(p_J))), \\ h \equiv_{\mathcal{P}} h' &\iff \sigma_{\mathcal{P}}(h) = \sigma_{\mathcal{P}}(h'). \end{aligned} \quad (3)$$

Read the tuple as the candidate’s observable fingerprint under this interface. Exact equality is reflexive, symmetric, and transitive, so it partitions the class. It does not assert that two candidates are the same mechanism or will agree on unseen inputs. The evaluation formed 36 signature classes and, in three A1 blocks, retained two classes. Specifically, the A0 interface used width 32, eight updates, periodic trajectories from single-center and alternating probes, and the ordered summaries final density and trajectory-mean density. Boundaries 0.125, 0.25, 0.5, 0.75, 0.875 define half-open bins with equality assigned to the upper-index bin; the resulting signed 16-bit bin codes and full specification header use one canonical byte encoding specified in the Supplement. The other A0 boundaries test simulator conformance but do not enter this signature. Pairwise “close enough” rules, midstream quantizer changes, or representative-only evaluation without full-interface equality would destroy the interpretation.

4.3 Distinguishing a bad winner from a bad language

Fix the loss, threshold τ , candidate set, class partition, look schedule, and coverage allocation before observing the evaluation outcomes. Let $\underline{L}_{h,s}$ be a lower bound on candidate loss at look s , with all such bounds valid on one

registered simultaneous event. For a signature class C , set $\underline{L}_{C,s} = \min_{h \in C} \underline{L}_{h,s}$. The entire language is declared inadequate only when

$$\text{MODEL_CLASS_INADEQUATE} \iff \min_C \underline{L}_{C,s} > \tau. \quad (4)$$

The inner minimum prevents a weakly bounded member from being hidden by a favorable class representative; the outer minimum asks whether even the most favorable registered class is above threshold. Thus $\tau = 0.2$ and a minimum class lower bound of 0.5, as in the constructed B1 blocks, satisfy the gate. The same display cannot be used if one candidate lacks a valid simultaneous bound. Missing execution, invalid deduplication, or an incomplete alpha ledger yields **INDETERMINATE**, not **MODEL_CLASS_INADEQUATE**.

The nested minima equal the minimum over all registered candidates, so the class layer preserves the weakest member and organizes reporting; it does not numerically strengthen the gate. In the current B0/B1 study the classes were singletons and the losses were exact complete-support values, so neither nontrivial class aggregation nor the repeated-look construction below drove a result.

4.4 Following error through time and across levels

Suppose F_t is a reference transition, $\hat{F}_t$ a candidate, and $x_{t+1} = F_t(x_t)$, $\hat{x}_{t+1} = \hat{F}_t(\hat{x}_t)$ form a paired path in one metric space (X, d) . On a declared admissible region, let F_t be L_t -Lipschitz and bound the local candidate discrepancy on that same path by $d(F_t(\hat{x}_t), \hat{F}_t(\hat{x}_t)) \leq \kappa_t$. Then

$$d(x_T, \hat{x}_T) \leq \left(\prod_{u=0}^{T-1} L_u \right) d(x_0, \hat{x}_0) + \sum_{t=0}^{T-1} \left(\prod_{u=t+1}^{T-1} L_u \right) \kappa_t. \quad (5)$$

The first term carries initial mismatch; the sum carries each later local error through the dynamics that follow it. Factors below one damp discrepancy, factors near one transmit it, and factors above one can amplify it. This is a deterministic pathwise statement. The bound is withdrawn if either trajectory leaves the stated region, if local error was bounded on a different path, or if the required metric or Lipschitz constant is unavailable. Benchmark C checked nine constructed cases; it did not prove control for physical or biological dynamics.

Cross-scale consistency asks a related but different question. Let $F_j : A_j \subseteq X_j \rightarrow X_j$ perform one update at level j , and let $\Gamma_j : B_j \subseteq X_j \rightarrow X_{j+1}$ coarse-grain it. Define the nonempty commutation domain

$$A_j^{\text{comm}} = \{x \in A_j \cap B_j : F_j(x) \in B_j, \Gamma_j(x) \in A_{j+1}\}.$$

The one-level defect is

$$\eta_\ell = \sup_{x \in A_\ell^{\text{comm}}} d_{\ell+1}(\Gamma_\ell F_\ell(x), F_{\ell+1} \Gamma_\ell(x)).$$

For paired routes started at $x \in A_\ell^{\text{comm}}$, assume every successive coarse image lies in the relevant registered commutation domain, both arguments compared through each later Γ_j lie in one pair-specific Lipschitz region, and Γ_j has finite constant G_j there. Then, for one fine-level update compared with one corresponding update at each coarser level,

$$\begin{aligned} & d_{\ell+r}(\Gamma_{\ell:\ell+r} F_\ell(x), F_{\ell+r} \Gamma_{\ell:\ell+r}(x)) \\ & \leq \sum_{q=0}^{r-1} \left(\prod_{j=\ell+q+1}^{\ell+r-1} G_j \right) \eta_{\ell+q}. \end{aligned} \quad (6)$$

The two sides are the two possible routes through the hierarchy: one update then repeated compression, or repeated compression with the corresponding single update at each level. The right side says that each local mismatch is weighted by the sensitivity of later coarse maps. This is a deterministic pathwise consistency bound, not a statistical validation theorem. A missing or empty commutation domain, type mismatch, false intermediate-domain premise, missing metric, or absent Lipschitz bound withdraws the cross-scale conclusion; a finite number cannot be invented to keep the workflow moving. Agreement within this bound is one conditional consistency test, not sufficient validation of the higher-level model.

4.5 Granting and challenging a distribution-conditional certificate

For iid calibration draws $W_i \sim Q$, let $K = \sum_{i=1}^m \mathbf{1}\{D(W_i) > \varepsilon_{\text{cert}}\}$ count violations of one prespecified coarse-graining event. With the map, event, tolerance, distribution, sample size, tail probability β , and target δ_{cert} fixed before calibration, the one-sided exact binomial construction of Clopper and Pearson [1934] gives the upper endpoint

$$U_{\text{CP}}(K, m; 1 - \beta) = \begin{cases} \text{Beta}^{-1}(1 - \beta; K + 1, m - K), & K < m, \\ 1, & K = m. \end{cases} \quad (7)$$

Calibration grants the certificate only if $U_{\text{CP}} \leq \delta_{\text{cert}}$. When $K = 0$, this endpoint is $1 - \beta^{1/m}$; with $\beta = \delta_{\text{cert}} = 0.05$, at least 59 zero-violation blocks are required. This 59-block statement applies specifically to the $K = 0$ design case; designs allowing $K > 0$ require a separate calculation. The formula is not a statement that violations are impossible. It is an exact distribution-conditional upper bound for one event under Q .

Role-separated held-out data answer a different question: is there enough evidence to withdraw the already granted certificate? With held-out count K_L , sample size m_L , and fixed level γ , compute

$$L_{\text{CP}}(K_L, m_L; 1 - \gamma) = \begin{cases} 0, & K_L = 0, \\ \text{Beta}^{-1}(\gamma; K_L, m_L - K_L + 1), & K_L > 0. \end{cases} \quad (8)$$

Withdrawal occurs if and only if $L_{\text{CP}} > \delta_{\text{cert}}$. Grant and withdrawal therefore have different directions and different evidential meanings: absence of withdrawal is not confirmation. Event changes, role reuse, tuning, pseudoreplication, distribution shift, or a broken certificate chain would make the calculation inapplicable.

These Clopper–Pearson statements are exact for the prespecified iid reference law. The successful role materializations were that law conditioned on within-role uniqueness and, for the held-out role, nonoverlap with calibration. Consequently the implemented-law coverage is bounded below by $1 - \beta - q_{\text{cal}}$ for calibration and $1 - \gamma - q_{\text{locked}}$ for held-out evaluation, where $q_{\text{cal}} = 1.74105158070704 \times 10^{-74}$ and $q_{\text{locked}} = 5.27842622087372 \times 10^{-74}$. These conservative total-variation deductions do not change any reported endpoint, interval, threshold, or status.

4.6 Paying for repeated looks

Let N and G be fixed candidate-slot and gate counts, $s \geq 1$ the look index, and $0 < \alpha < 1$ the global budget. One valid allocation is

$$\alpha_{hgs} = \frac{6\alpha}{NG\pi^2 s^2}, \quad r_{hgs} = \sqrt{\frac{\log(2/\alpha_{hgs})}{2n_s}}, \quad (9)$$

The s^{-2} schedule is used because its infinite sum is finite: across all reserved candidate–gate–look streams, the allocated error is at most α . This is the mathematical reason an early look and a later look cannot both spend the full budget. Candidates revealed adaptively need prespecified summable weights before their outcomes are inspected. Unplanned looks, missing weights, dependence without an appropriate replacement bound, or an incomplete ledger void the affected elimination.

Here n_s is the cumulative sample size available at look s ; replacing it by s is valid only for a schedule that explicitly enforces $n_s = s$. This post-result formal correction was established by call-graph audit to be nondependent: no protected endpoint called this repeated-look radius, and current B0/B1 used exact complete-support losses.

4.7 Deciding when lower-level precision is enough

For a prespecified fidelity comparison $k \prec k'$, define

$$O_{j,k}(X) = O_j(S_k(X)_{0:T_j}), \quad \Delta_j(k, k') = \text{Agg}_j[d_j(O_{j,k}(X), O_{j,k'}(X))].$$

The observable, metric, tolerance $\varepsilon_j^{\text{fid}}$, horizon, input or environment law, aggregation rule, paired blocks, simultaneous uncertainty procedure, and decision map are all fixed in advance. Fidelity k is sufficient only when

$$\text{Sufficient}(k) \iff [\forall k' \in \mathcal{R}(k), \forall j : \Delta_j(k, k') \leq \varepsilon_j^{\text{fid}}] \wedge [\forall k' \in \mathcal{R}(k) : \mathcal{D}(k') = \mathcal{D}(k)]. \quad (10)$$

The first bracket protects upper-level quantities; the second protects the scientific decision. Sufficiency is relative to the declared refinements $\mathcal{R}(k)$, not to arbitrary future precision or exact real arithmetic. A tolerance breach, signature-boundary contact, changed certificate state, decision reversal, failed invariant, missing output, or incomplete refinement prevents a sufficiency decision.

For stochastic or quantum observables, each comparison uses a registered simultaneous confidence set or certified error interval, not a point estimate alone. The set must lie wholly within one quantization cell and inside the fidelity tolerance; contact with a cell boundary returns `SIGNATURE_INDETERMINATE`, and its mislabel probability is charged to the registered family error ledger.

5 Relationship to existing methods

Program search, algorithmic theories, computational-universe proposals, and universal induction all predate APHFS [Zuse, 1969, Fredkin, 1990, Wolfram, 2002, 2020, 't Hooft, 2016, Schmidhuber, 2000, Hutter, 2005]. Feynman’s quantum-simulation proposal and Deutsch’s universal quantum computer likewise predate this framework; neither establishes completeness of a candidate search [Feynman, 1982, Deutsch, 1985]. Version-space search and candidate elimination long predate this project [Mitchell, 1977, 1982]. Program sketching and syntax-guided synthesis likewise formalize finite grammar-constrained construction and verification, including the possibility that a grammar is unrealizable for a specification [Solar-Lezama et al., 2006, Alur et al., 2013, Kim et al., 2023]. DreamCoder, symbolic regression, and sparse equation discovery show other ways to search structured programs or equations [Ellis et al., 2021, Bongard and Lipson, 2007, Schmidt and Lipson, 2009, Brunton et al., 2016, Udrescu and Tegmark, 2020]. APHFS does not claim priority for automated discovery or computational descriptions of nature.

Quantum cellular automata (QCA) and discrete quantum dynamics are direct predecessors of the long-term pre-particle program. Early work formalized one-dimensional QCA and quantum lattice gases [Watrous, 1995, Meyer, 1996]; local unitary lattice dynamics were also used to recover Weyl-, Dirac-, and Maxwell-like continuum behavior under stated assumptions [Białynicki-Birula, 1994]. Schumacher and Werner formalized reversible QCA with finite propagation; localizability and one-dimensional index results further developed the structure [Schumacher and Werner, 2004, Arrighi et al., 2011, Gross et al., 2012], with later overviews describing the field [Arrighi, 2019, Farrelly, 2020]. In particular, D’Ariano and Perinotti and related work derive Dirac-like large-scale dynamics from QCA satisfying explicit unitarity, locality, homogeneity, and discrete-isotropy premises [D’Ariano and Perinotti, 2014, Bisio et al., 2015, 2016, D’Ariano and Perinotti, 2017]. APHFS claims no priority for, and neither reproduces nor validates, those physical derivations. Its distinct target is the accounting and failure discipline under which candidate narrowing, residual ambiguity, whole-class inadequacy, and insufficient evidence would all remain reportable.

Finite-class learning, sieve estimators, structural risk minimization, and complexity penalties motivate the risk analysis [Grenander, 1981, Geman and Hwang, 1982, Vapnik, 1995, Barron and Cover, 1991]. Partial identification and model-confidence-set methods make set-valued conclusions explicit in other statistical settings [Manski, 2003, Hansen et al., 2011]. Work on misspecification, posterior predictive criticism, M-open analysis, and open-set recognition makes clear that a supplied class can be wrong [White, 1982, Gelman et al., 1996, Yao et al., 2018, Scheirer et al., 2013]. APHFS specializes that concern to an executable grammar in which a whole-class decision depends on simultaneous candidate-level bounds and complete failure accounting.

Systems biology has a mature literature on structural and practical parameter identifiability, profile likelihood, observability, and sloppy sensitivity directions [Bellman and Åström, 1970, Gutenkunst et al., 2007, Raue et al., 2009, Chis et al., 2011, Villaverde et al., 2016]. Those analyses ask whether parameters or parameter combinations are determined under a model and observation design. An APHFS canonical signature instead labels an entire executable candidate under one fixed probe, quantizer, and serialization interface. The neighboring ideas should not be conflated: an operational signature class can contain different mechanisms, while parameter identifiability can fail within one mechanism. APHFS also permits a separately justified rejection of the whole declared grammar, but the present ECA study does not show empirical superiority over established biological identifiability tools.

Time-uniform confidence sequences address repeated evaluation [Howard et al., 2021, Waudby-Smith and Ramdas, 2024]; the displayed APHFS schedule instead uses a prespecified summable union allocation. Multifidelity and cost-aware methods address allocation under limited resources [Babu et al., 2004, Jamieson and Talwalkar, 2016, Li et al., 2018, Kandasamy et al., 2017]. Goal-oriented error estimation controls quantities of interest, including in hierarchical and lattice settings [Prudhomme and Oden, 1999, Oden et al., 2005]; APHFS’s decision-relative fidelity interface is an integration with candidate and failure accounting, not a priority claim for that idea. Cellular- automaton coarse-graining and its relation to predictability also have direct precedents [Israeli and Goldenfeld, 2004, 2006]. Projection and predictive-state traditions show complementary aspects of coarse description: projection-operator formalisms show how eliminating unresolved variables can produce memory kernels [Zwanzig, 1961, Zwanzig et al., 1972, Mori, 1965], while predictive-state and molecular-kinetics traditions motivate observable-grounded state representations [Littman et al., 2001, Prinz et al., 2011]; statistical complexity and information measures offer related diagnostics [Crutchfield and Young, 1989, Hoel, 2017, Rosas et al., 2020]. The contribution evaluated here is not a new theorem in any one of these areas. It is the disciplined connection among them: a search is not complete without accounting, a winner is not adequate without a class test, a coarse description is not reliable without domain and memory conditions, and a held-out result is not outcome-blind if the pipeline can be retuned after exposure.

No broad method family is intrinsically restricted to one winner or incapable of abstention. Version spaces, confidence sets, partial identification, model criticism, synthesis unrealizability, and identifiability analysis already provide related set-valued or failure-aware ideas. The Supplement therefore compares concrete procedures and assumptions rather than assigning generic Yes/No capabilities. APHFS-specific content lies in the proposed integration of executable address accounting, operational signatures, complete-bound class decisions, decision-relative fidelity, failure-preserving execution, and role-separated evaluation. This targeted review did not establish historical priority for that integration. Any AIC, BIC, SINDy, or other baseline comparison requires a new prespecified study and new evaluation roles.

6 Future fault-tolerant quantum execution interface

APHFS is hardware-agnostic at the level of scientific logic. Its present implementation is classical because a complete finite grammar makes candidate semantics, accounting, failure, and held-out evaluation directly inspectable. A future fault-tolerant quantum execution layer would not relax those requirements. It would change which candidate state spaces and dynamical refinements can be executed, while adding state-preparation, operator-access, error-correction, measurement, and classical-post-processing obligations. This is a roadmap; no quantum implementation, resource estimate, or advantage was tested.

Marked-item search is not complete accounting. Grover-type algorithms reduce oracle queries for locating a marked item in an unstructured search, with oracle-model limits characterized by Bennett, Bernstein, Brassard, and Vazirani [Grover, 1996, Bennett et al., 1997]. APHFS calls a run exhaustive only when every addressable candidate receives a terminal status. Producing N such records has an output burden of order N , so a complete terminal ledger does not inherit the square-root query count of a one-item search. Quantum search may help a separately declared task such as locating one counterexample; it cannot turn N required ledger entries into $\sqrt{N}$ entries.

What quantum hardware may change. A prospective role is executing quantum-native candidate dynamics—local unitary circuits, quantum walks, QCA, Hamiltonians, or channels—for which efficiently specified preparation, evolution, and measurement interfaces exist. Quantum simulation algorithms can make particular encoded dynamics accessible [Lloyd, 1996, Low and Chuang, 2019], but advantage is specific to the grammar, instance, observable, access model, accuracy target, and classical baseline. Preparation, oracle or block encoding, error correction, sampling, and readout may dominate [Plesch and Brukner, 2011, Fowler et al., 2012, Gidney and Ekerå, 2021]; stabilizer, tensor-network, or other classical structure may remove an apparent advantage. The paper therefore

makes no generic polynomial-versus-exponential claim.

For a future quantum candidate language, the ledger would use a typed nested record rather than a heterogeneous numerical vector:

```
QuantumResourceRecord = {semantics, input_and_preparation,
                          operator_access, logical_circuit, qec_mapping,
                          physical_estimate, measurement_and_readout,
                          classical_postprocessing, error_budget, provenance}.
```

The logical circuit fixes a native gate basis and declares whether Toffoli is primitive or compiled into T gates, so the two counts cannot be charged twice. The QEC mapping records code family and distance, physical noise assumptions, decoder, factory, layout/connectivity, cycle time, and allocated logical-failure probability; a physical-qubit range is a derived estimate, not an alternative field beside the QEC model. Preparation and operator access state the input’s origin, normalization, construction cost, target error, success probability, and reusability. Measurement records observables, estimators, shots, simultaneous intervals, guard bands, and readout; classical verification, output materialization, storage, and wall time remain billable. Different units are compared componentwise by default. Scalarization is permitted only through a prespecified cost map, and physical estimates must carry their uncertainty. Instance-level values require a separate resource study; none is asserted here. Fault-tolerant chemistry studies illustrate why end-to-end accounting matters [Reiher et al., 2017].

Sampling, signatures, and error semantics. Quantum shots estimate observables; they do not reveal exact hidden values. In the proposed extension, a prespecified simultaneous interval must lie wholly inside one fixed quantization cell before that cell receives a canonical label. Contact with a guard band returns `SIGNATURE_INDETERMINATE`; exact equality of assigned labels then remains a partition among determinate candidates. This is a roadmap construction, not a tested theorem. The quantum error ledger distinguishes preparation, encoding/access, algorithmic simulation, gate synthesis, logical/QEC failure, measurement, classical processing, and coarse-graining error. Only terms in a common probability or metric may be combined. Without a registered logical-failure model or valid bound, the affected decision is `INDETERMINATE`; no universal independent-gate formula is assumed. Repeated state preparation and shots remain possible, so the no-retry rule continues to be an inferential safeguard against outcome-dependent tuning, not a consequence of no-cloning.

Detailed Q0–Q5 milestones and the full schema appear in the Supplement. Candidate classes must state their locality, causality or no-signalling, symmetry, unitarity/CPTP, conservation, and continuum-limit assumptions. Bell-type violations constrain local hidden-variable accounts under the theorem’s assumptions; they do not require every pre-particle grammar to be QCA. Compute expands which dynamics can be evaluated; evidence still determines what can be justified.

7 Study design

Elementary cellular automata (ECA) provide a deliberately small stress test. Each candidate is a binary radius-one rule: eight neighborhood inputs map to eight output bits, so exactly 256 rule tables exist. The grammar is rich enough to produce collisions, ambiguity, non-closure, memory, and policy trade-offs, yet small enough that a separate implementation could enumerate every rule. It tests APHFS machinery, not a physical or biological model.

The grammar, endpoints, observation units, thresholds, fidelity contracts, cost ledger, and failure semantics were fixed before evaluation. Development constructed the pipeline; calibration was used only for the Rule 170 D2-CERT event; and a separate held-out role evaluated the completed pipeline once, without retuning or retry. For blockwise endpoints, one role-derived block—not a cell or time point—was the observation unit. The ideal reference law sampled blocks independently; successful role materialization conditioned that law on uniqueness and cross-role nonoverlap, as quantified with the exact-binomial formulas above. The protocol and artifacts were pre-specified and

internally hash-committed before execution according to the project record. The disclosed hashes bind content and make later alteration detectable. No third-party trusted timestamp or public preregistration preceded execution, so the original chronology is documented internally but is not independently time-verified.

Six benchmark families asked whether all rules were accounted for and a generating rule could be retained without forcing uniqueness (A); whether an in-class control was retained and a deliberately distinguishable radius-two class was rejected (B); whether path-error bounds were implemented correctly (C); whether full-state identity, density non-closure, certificate withdrawal, and memory repair were distinguished (D); how four resource policies compared on the same observations and cost definition (E); and whether project-defined Rule 54/110 trajectory fixtures were recognized (F0). Sample counts, thresholds, fidelity tiers, and terminal actions are given in the Supplement.

A0, C, D0, D1, and D2-MEM are analytical, deterministic, or formula-conformance checks. A1, B0, B1, E, and F0 use 64 held-out blocks to audit constructed workload behavior; their blockwise intervals describe adverse implementation or accounting events under the reference role law, not general model-discovery accuracy. D2-CERT audits certificate machinery on a provably closed map. Missing output, timeout, numerical failure, fidelity instability, or incomplete accounting could not be assigned a favorable value.

8 Results

The evaluation separated three conclusions that a leaderboard would conflate: retention without unique identification, rejection of an inadequate class, and non-withdrawal of a certificate without confirmation (Figure 3). The results below distinguish an analytical statement, deterministic or formula conformance, constructed-workload conformance, a role-separated held-out machinery check, a post-hoc operating characteristic, and a future roadmap. These categories answer different questions and are not pooled into one empirical performance score.

8.1 Complete accounting exposed ambiguity rather than erasing it

All 256 ECA rule IDs received a terminal ledger entry. The A0 truth-table and paired production/reference trajectory census across four specified boundaries—periodic, fixed zero, fixed one, and edge-replicate (historical code token: `reflect`)—produced 0/256 discrepancies and formed 36 canonical signature classes. The finite language was fully accounted for under that interface; enumeration beyond the ECA grammar is outside this claim. The 36 classes are operational classes induced by the finite probes, quantizers, and serialization used here. They are neither universal dynamical-equivalence classes of ECA rules nor a standard symmetry-class count. Complete accounting supports a statement about coverage of this language; it does not establish that the language is adequate.

The A1 result was not merely “the generating rule was found.” The data-generating rule and its signature class remained in the retained set in all 64 blocks, with 0 adverse blocks. Yet 59 blocks retained one rule and five retained two; 61 retained one signature class and three retained two. The finding is set-valued identification under finite observations. It argues for more discriminating observables when uniqueness matters, rather than a hidden tie-break. The alternatives are operationally indistinguishable here, not ontologically identical.

For a correct implementation, zero A1 adverse blocks follow from generating each noiseless history with a member of the enumerated grammar and retaining every candidate with exact loss zero. The 64-block interval therefore characterizes adverse implementation/accounting events under the reference role law; it is not an estimate of open-world discovery accuracy. The real finding remains the set-valued ambiguity that the finite observation interface exposed.

8.2 The class gate separated a bad model from a bad language

The paired B endpoints tested opposite constructed cases. In B0, all 64 complete-support in-class blocks retained the class containing the generating rule. In all 64 complete-support blocks from the deliberately distinguishable radius-two construction, the ECA class returned `MODEL_CLASS_INADEQUATE`. Every minimum signature-class lower

What the evaluation resolved and what remained ambiguous

Panels separate immutable outcomes from their scientific evidence classes.

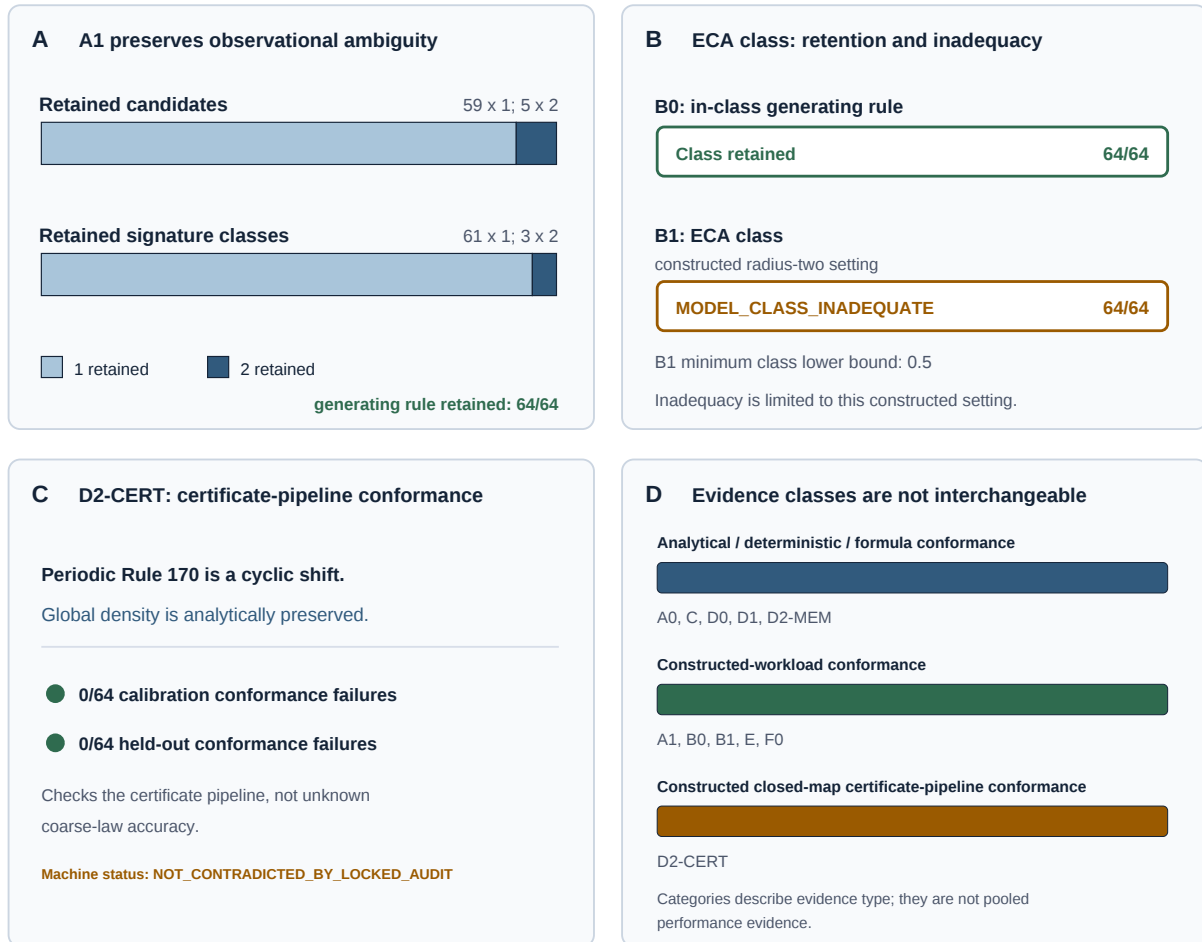


Figure 3: Scientifically distinct outcomes and evidence classes. A1 preserves set-valued ambiguity; B0 and B1 contrast in-class retention with one deliberately distinguishable constructed out-of-class case; D2-CERT exercises certificate grant and withdrawal machinery on the analytically density-preserving periodic Rule 170 map. The final panel separates analytical/deterministic checks, constructed-workload conformance, and constructed closed-map certificate-pipeline conformance; these categories are not pooled into a model-discovery performance score.

bound was 0.5, above the threshold 0.2. The corresponding B0 minimum was zero, so no in-class block produced false inadequacy.

The gate thus distinguished this constructed out-of-class setting from its in-class control. B1’s outer-XOR generator guarantees disagreement with every radius-one ECA candidate on half of its complete 32-row support, so the observed 64/64 decision is a conformance check of exact-loss, complete-ledger, and gate machinery. B1 does not estimate power against arbitrary misspecification, and the endpoint-wise block interval does not turn it into such an estimate.

8.3 Error and coarse-graining claims remained conditional on their interfaces

All nine path-bound cases completed with no violation or domain exit. The largest raw reference–candidate discrepancy was 0.1662, meaningful only in relation to its stated bound. The smallest margin was near -10^{-16} and remained within the fixed 10^{-12} tolerance, so it did not change a decision. These are constructed implementation checks; full-precision values remain in the Supplement and released source data. The tolerance governs the comparison at the numerical boundary; it does not turn the nine cases into a general trajectory-accuracy guarantee.

The deterministic coarse-dynamics checks reproduced Rule 204 full-microstate identity, the Rule 90 density non-closure witness, and exact lag-1 repair in the two-bit memory system. D2-CERT has a narrower, now explicit evidence class: constructed closed-map certificate-pipeline conformance. Under periodic boundaries Rule 170 copies the right neighbor, hence performs a cyclic shift and preserves global binary density exactly. With the registered identity predictor and zero tolerance, a correctly executed block cannot violate the event. Calibration and held-out evaluation each observed 0/64 violations, checking role separation, simulator/fidelity agreement, and the grant/withdrawal pipeline for this construction rather than estimating uncertain coarse-law accuracy.

The immutable held-out two-sided 95% interval was $[0, 0.0560]$, the one-sided lower endpoint was zero, and the withdrawal rule did not fire. These exact-binomial values describe a hypothetical Bernoulli rate of block-level implementation/conformance failure under the iid reference law; they are not uncertainty about the analytical density identity itself. The exact machine status remains `NOT_CONTRADICTED_BY_LOCKED_AUDIT`. It neither confirms a broader closure theorem nor grants a new certificate, and it does not transfer to another boundary, coarse variable, or physical system.

A post-hoc operating-characteristic calculation, using the unchanged $m_L = 64$, $\gamma = \delta_{\text{cert}} = 0.05$ rule, shows that withdrawal first occurs at $K_L = 7$: $L_{\text{CP}}(6, 64) = 0.04162$ and $L_{\text{CP}}(7, 64) = 0.05247$. Its withdrawal probability is 0.461 at a hypothetical implementation-event rate of 0.10 and 0.864 at 0.15. This calculation is not a registered endpoint and does not change the exact D2 status. Because a correct implementation cannot violate the registered construction, the grid describes operating characteristics of the withdrawal machinery under hypothetical faults, not uncertainty about the Rule 170 density identity.

All prespecified fidelity comparisons preserved their upper-level observables and decisions and required no further refinement. This conclusion is specific to their observables, tolerances, horizons, and input laws. The comparisons distinguish lower-level numerical change from change that matters to the stated scientific inference.

8.4 One prespecified order was least costly on this workload

Both `fixed_order` and `development_frozen_order` were distinct permutations of the same 256 candidates generated before protected execution by sorting salted SHA-256 digests; neither carried a scientific ordering prior. All four policies matched the exhaustive final decision, best valid candidate, and failure status in all 64 paired blocks. Their costs nevertheless differed because the role-derived generating rules occupied different positions in those fixed permutations: `fixed_order` used 75,080 units, development-frozen order 89,816, adaptive fidelity 92,592, and exhaustive search 132,096. Relative to exhaustive search, the first three saved 43.16%, 32.01%, and 29.91%, respectively.

Adaptive fidelity cost about 23.32% more than `fixed_order` and 3.09% more than `development_frozen_order`. Its cheap tier evaluates four support rows for all 256 candidates before full-tier evaluation and reference refinement;

Policy cost and its composition

Fixed order was least costly; adaptive fidelity did not win on this workload.



Components sum to each total; all policies matched the exhaustive decision in 64/64 blocks.

Figure 4: Benchmark E costs on the same 64 observation streams. Stacked components include candidate-block steps, ordering, probes, reference refinements, and retry/other costs; they sum to each reported total. All policies agreed with exhaustive search, but one prespecified salted pseudo-random order was least costly and adaptive fidelity did not win this constructed workload.

that overhead was not repaid in this grammar. The supported conclusion is narrow: one prespecified early-stopping order was least costly on this constructed workload and registered cost map, while adaptive fidelity did not win. Cost units are bookkeeping components, not universal wall-clock time, and the ranking does not generalize to multifidelity search.

8.5 Known-fixture recovery completed the finite-grammar test, not the biological bridge

F0 contained 32 clean translations, 16 single-cell perturbations, and 16 negative controls from project-defined Rule 54/110 whole-trajectory fixtures. The rules are established ECA fixtures [Cook, 2004, Martínez et al., 2006], but the exact seeds and patterns used here were defined by this project rather than reproduced from literature. Similarity was 1.000 for clean translations, 0.9988 for perturbations, and 0.511–0.554 for negative controls. This passed as known-template conformance: recognition of project-defined fixtures and controls, not persistent-structure discovery, prediction, velocity or lifetime estimation, or a novel phenomenon. The 32/16/16 counts are descriptive realized strata of the prespecified iid mixture, not additional inferential endpoints, and the comparison concerns whole trajectories only. Because positive cases are generated from the same registered templates and negative cases from a project-defined digest construction, the zero adverse blocks are constructed-workload conformance for a correct matcher, not an estimate of general pattern-discovery performance.

9 Discussion

9.1 Honest outcomes are part of the method

APHFS preserved outcomes that a success-only pipeline could collapse. A1 retained the generating rule while admitting residual ambiguity. The constructed B1 class was declared inadequate only after every member had a valid bound. D2-CERT’s constructed closed-map pipeline reached its exact non-withdrawal state without implying broader confirmation. Each outcome motivates a different response: collect sharper observations, specify a broader language in a new study, or retain a conditional statement.

Failure semantics define the same boundary from the other side. An incomplete ledger or invalid simultaneous bound suspends the class decision; a domain exit withdraws the cross-scale bound. Rule 90 density non-closure, lag-1 memory repair in the two-bit system, and the distinct Rule 170 certificate interface show why coarse descriptions must be judged by their own maps, domains, and questions. This separation turns apparent failure into experimental guidance. Residual ambiguity calls for more discriminating observables; class inadequacy calls for a newly specified language; and an indeterminate outcome calls for repairing evidence or computation before making a scientific claim.

9.2 Greater complexity is not automatically greater efficiency

Benchmark E supplies a concrete caution: adaptive fidelity saved resources relative to exhaustive search, but one salted pseudo-random early-stopping order was cheaper while reaching the same decisions. The two static orders differed only by prespecified salts, not by a learned scientific prior. Adaptation helps only when information gains exceed overhead. This constructed comparison supports neither a universal fixed ordering nor a rejection of multifidelity methods.

9.3 Conditions under which APHFS would lose practical value

APHFS would lose practical value if it abstains so often that it cannot guide the next measurement; if, across registered domain studies, its decisions repeatedly match a much cheaper baseline without compensating evidential benefit; if reasonable probe changes make the operational signature partition unstable; if candidate languages

repeatedly fail every domain bridge without producing informative revisions; or if the declared resource burden makes evaluation infeasible. These are failure criteria for the usefulness of the framework, distinct from failure of one candidate model. They require future comparative studies rather than reinterpretation of the current locked blocks.

9.4 The next registered biochemical bridge

The next study can be publicly preregistered around a finite enzyme-kinetics grammar that separates mechanistic mass-action systems from reduced rate laws. Its in-class language would include bounded Michaelis–Menten, Hill/cooperative, competitive- and noncompetitive-inhibition, substrate- inhibition, delay, and allosteric candidates, with equations, units, parameter domains, initial conditions, conservation constraints, and semantic duplicates fixed. A new data-generating law would define perturbations, substrate and inhibitor levels, sampling times, replicates, batches, noise, censoring, missingness, and detection limits. Partial steady-state and time-series concentrations would then test operational non-identifiability, while an explicitly excluded but distinguishable mechanism could provide a prespecified class-inadequacy challenge.

The study would define a recovery- or homeostasis-relevant upper observable after substrate or inhibitor perturbation. Solver tolerance, time step, state resolution, and horizon refinements would be accepted only when that observable and its retain/inadequate/indeterminate decision remain stable. New development, calibration, and held-out roles would be generated for this study; the current ECA roles would not be reused. The new protocol would define sample size, power, multiplicity, stopping, and fair cost accounting before evaluation. AIC, BIC, profile likelihood, top-1 selection, or SINDy could enter only as prespecified empirical baselines under the same observations and budgets. Ambiguity, class inadequacy, fidelity failure, and policy non-advantage would remain admissible rather than being recoded as success. This blueprint is a future study, not biochemical evidence in the present one.

Repeated bridges of this kind are required before movement toward cellular organization. Whole-cell models illustrate both integrative reach and demanding validation [Karr et al., 2012, Thornburg et al., 2022, 2026a,b]. Aging becomes testable only when models predict longitudinal homeostatic drift and recovery under suitable data [López-Otín et al., 2013, 2023]; rejuvenation follows later through causal, safety-constrained intervention studies. More classical or quantum compute may expand the evaluation, but it cannot substitute for domain evidence; no quantum advantage was tested here.

10 Limitations

The study is deliberately narrow. Completeness applies only to the 256 radius-one binary rule addresses under the specified interfaces. A0 is internal implementation conformance; C, D0, D1, and D2-MEM are analytical, deterministic, or formula checks. A1, B0, B1, E, and F0 are constructed workload-conformance endpoints whose blockwise intervals concern adverse implementation or accounting events under the reference role law, not general discovery accuracy. B1 is deliberately distinguishable and F0 uses project-defined fixtures. D2-CERT checks certificate machinery on an analytically closed Rule 170 density map; its exact status is non-withdrawal, not confirmation. Endpoint-wise exact-binomial intervals are not a simultaneous family-wide guarantee, and their exactness applies to the iid reference law with the disclosed conditioning correction for the implemented roles.

The scalar reference and vectorized production paths were written separately but within the same project with substantive AI assistance; their agreement is not external replication. Benchmark E’s ranking is specific to two salted pseudo-random orders, one constructed workload, and one cost map. No quantum interface was implemented, and no external particle, chemical, molecular, cellular, aging, clinical, or intervention data were analyzed. The present contribution is a methodological systems-biology architecture and finite ECA machinery test, not a validated domain model or therapeutic claim.

11 Data and code availability

The accompanying Supplement and reproducibility archive provide the executable source tree, fixed configurations and schemas, environment locks, safe aggregate source tables, figure-generation code, A0 candidate/signature records, and read-only tools for reconstructing reported endpoint counts, exact-binomial calculations, aggregate tables, figures, intervals, and costs. The same materials are available in the APHFS repository at <https://github.com/cmy515006420/APHFS>. The archive and repository do not include raw calibration or locked role values or the protected result container; they do not rerun the original one-time evaluation and are not an independently written implementation, new-role replication, or validation on new data. Code is licensed under the MIT License; the manuscript, documentation, figures, and released aggregate and source data use CC BY 4.0.

12 Declarations

Intellectual origin and author responsibility. Mingyuan Chen conceived the APHFS research program and its central thesis for this project, selected the scientific questions and claim boundaries, approved the study design, and made the final scientific judgments. AI systems contributed substantive methodological, formal, software, and editorial assistance as disclosed below; the author decided what to adopt and accepts responsibility for the work. This origin statement records authorship of the present APHFS integration, not historical priority for its established components, automated discovery, or computational descriptions of nature.

Author contributions. Mingyuan Chen: Conceptualization, Methodology, Formal analysis, Validation, Investigation, Data curation, Visualization, Writing—original draft, Writing—review & editing, and Project administration.

Funding. No external funding was received for this work.

Competing interests. The author declares no competing interests.

Ethics statement. Not applicable. This theoretical and computational study used no human participants, human data, animals, clinical intervention, or identifiable private data.

Acknowledgments. None.

AI-assisted technologies. The author used OpenAI ChatGPT (primarily GPT-5.6 Sol through the standard and reasoning settings available in the author’s account) and OpenAI Codex for outlining and drafting, methodological and formal suggestions, software implementation and refactoring, tests, mathematical exposition, figure preparation, literature organization, adversarial review, release engineering, and editing. Anthropic Claude (primarily Claude Opus 5 and Claude Fable 5.1) was used for methodological critique, adversarial review, and revision suggestions. Google Gemini and xAI Grok were used only for limited exploratory critique; their exact versions were not recorded, and no specific retained substantive contribution is attributed to them. The model-version history is therefore partial.

AI outputs were treated as provisional suggestions, not evidence or independent validation. Mingyuan Chen originated the central research direction, set the scientific questions and claim boundaries, reviewed the manuscript and reported results, checked the functions and assumptions of Equations (1)–(10), the Clopper–Pearson calculations, relevant portions of load-bearing sources, reported file-integrity checks, and directly observed the public read-only aggregate recomputation. The author has functional-level understanding of the core scientific code but did not conduct a repository-wide line-by-line review. The author made the final scientific decisions and accepts responsibility for the work within this disclosed review scope. No human, animal, clinical, or identifiable private biological data were supplied to these systems. AI systems are not authors or CRediT contributors.

13 Conclusion

APHFS is a discipline for moving from imagination through executable mathematics to evidence. Bottom-up candidate generation meets top-down reality constraints; complete accounting within a finite grammar leaves room for ambiguity, whole-class inadequacy, and an indeterminate result. Coarse-graining is not accepted prematurely, and fidelity is sufficient only when refinements preserve upper-level observables and decisions. Compute expands evaluation; evidence constrains justification. A future fault-tolerant quantum layer could expand which pre-particle or quantum-native dynamics are executable, but it would neither remove the output burden of complete terminal-ledger accounting nor supply empirical support. Biology, aging, and rejuvenation remain downstream research destinations.

In the complete 256-rule ECA testbed, the generating rule was retained but was not always unique; the constructed B1 setting rejected an inadequate class while B0 retained its in-class control; and the provably density-preserving Rule 170 construction exercised certificate machinery without withdrawing or confirming a broader claim. All policies agreed with exhaustive search, yet one prespecified salted pseudo-random order was least costly on this workload and adaptive fidelity did not win. Ambiguity, constructed scope, and a negative method comparison are part of the result.

The next bridge should test a small biochemical model class with domain observations collected independently of model development, new role-separated data, protected recovery observables, and the same permission to fail. Only repeated empirical bridges could justify movement toward cellular models, aging as loss of resilience, and eventually safety-constrained rejuvenation research.

A Supplementary derivations, protocol, and immutable results

This supplement provides derivations, the complete frozen protocol, safe aggregate outcomes, and provenance commitments for the accompanying APHFS preprint. It reports only safe aggregates and commitments read from the immutable first-and-only locked result and source data checked by independent read-only recomputation. The result bundle SHA-256 is 6fb3f08e48b4d3496e190fbc38b029ee7c15e327c504924f8c03b6e2083aec9c, and the execution-receipt SHA-256 is 730bfe099a74b1ccfa281e653e9330bf91a5e73c30b24a58cf52dd88028c4766. No raw calibration or locked role value is reproduced here, and no development or public-mock output is relabeled as scientific evidence. The locked result was not rerun, replaced, tuned, or used to add a result-dependent endpoint.

A.1 Finite-class derivation and scheduling

For fixed h , Hoeffding’s inequality gives [Hoeffding, 1963]

$$\Pr\left(\left|\widehat{R}_n(h) - R(h)\right| > u\right) \leq 2e^{-2nu^2}.$$

For the fixed single-valued parser and address set $\mathcal{A}_b = \bigcup_{j=0}^b \{0, 1\}^j$, the executable semantic image obeys

$$N_{b,r} = |\mathcal{H}_{b,r}| \leq |\mathcal{A}_b| = 2^{b+1} - 1 < 2^{b+1}.$$

The ledger itself is address-indexed: invalid and resource-infeasible addresses receive terminal rows, while parser aliases remain separate unless a canonical-address rule was fixed in advance. Thus complete accounting is relative to the declared parser/address policy, not to all semantic models or nature. Therefore the union bound is

$$\Pr\left(\max_{h \in \mathcal{H}_{b,r}} \left|\widehat{R}_n(h) - R(h)\right| > u\right) \leq 2N_{b,r}e^{-2nu^2}.$$

Setting

$$c_b(\delta_{\text{conf}}) = \sqrt{\frac{(b+2)\log 2 + \log(1/\delta_{\text{conf}})}{2n}}$$

makes the right-hand side no greater than δ_{conf} . The optimizer is required to satisfy

$$\widehat{R}_n(\widehat{h}_n) + \lambda_n L_d(\widehat{h}_n) \leq \min_{h \in \mathcal{H}_{b,r}} \{\widehat{R}_n(h) + \lambda_n L_d(h)\} + \xi_n.$$

The three-line comparison in Theorem 1 uses $L_d(h_{b,r}^*) \leq b$, $-L_d(\widehat{h}_n) \leq 0$, and $\lambda_n \geq 0$. Because $\mathcal{H}_{b,r}$ is explicitly assumed nonempty and finite, its risk minimum is attained; no infimizing-sequence step is needed. The nonempty registered envelope, common measurable $[0, 1]$ -valued loss interface, and common evaluation distribution are all fixed before and without using the evaluation sample; that same loss and distribution define every displayed risk. The inclusion $\mathcal{H}_\infty \supseteq \mathcal{H}_{b,r}$ makes $\varepsilon_{\text{app}}(b, r) = R_{b,r}^* - R_\infty^*$ nonnegative. The resource vector r restricts which descriptions execute, while the displayed cardinality exponent depends only on the bit budget b .

A sufficient prespecified scheduling regime makes every term on the right side of Equation (2) vanish: $b_n \rightarrow \infty$, $\varepsilon_{\text{app}}(b_n, r_n) \rightarrow 0$, $b_n/n \rightarrow 0$, $\delta_{\text{conf},n} \rightarrow 0$, $\log(1/\delta_{\text{conf},n})/n \rightarrow 0$, and $\lambda_n b_n + \xi_n \rightarrow 0$. This yields convergence in probability relative to the declared envelope comparator, not natural truth risk or model identification. Almost-sure convergence additionally needs a summable failure schedule $\sum_n \delta_{\text{conf},n} < \infty$. The class, loss, distribution interface, and resource schedule are fixed before each evaluation sample; the conclusion does not cover adapting a grammar to the same evaluation data. The schedule need not be nested for the in-probability statement. A separate $b^{-\nu}$ rate illustration would additionally require a nested or otherwise monotone approximation structure that validates the approximation law along the schedule.

Table 2: Notation register for the formal interfaces. Symbols are scoped so that a confidence level, certificate tolerance, fidelity tolerance, and look index cannot silently exchange meanings.

Symbol	Meaning	Scope boundary
$a \in \mathcal{A}_b$	Declared ledger address	Not the semantic candidate itself.
r	Classical execution-resource envelope	Distinct from the repeated-look radius r_{hgs} .
δ_{conf}	Theorem 1 tail-failure probability	Not a certificate target.
$\varepsilon_{\text{app}}(b, r)$	Approximation gap to the declared comparison envelope	Not a certificate or fidelity tolerance.
κ_t	Paired-path local discrepancy in Equation (5)	Defined at $\hat{x}_t$ in the registered metric and domain.
η_ℓ	One-level cross-scale commutation defect	Conditional on typed maps and admissible intermediate images.
$\varepsilon_{\text{cert}}$	Coarse-certificate event tolerance	Fixed with the event and reference law before calibration.
δ_{cert}	Certificate grant/withdrawal probability target	Not the theorem tail probability.
$\varepsilon_j^{\text{fid}}$	Observable-specific fidelity tolerance	Relative to one protected observable, horizon, input law, and refinement family.
s, n_s	Registered look index; cumulative observations at that look	Only s enters alpha spending; only n_s enters the radius denominator.
$q_{\text{cal}}, q_{\text{locked}}$	Total-variation bounds induced by role acceptance	Not quantizers; quantizers retain the notation q_j only inside Equation (3).

A.2 Path-error induction

Let $e_t = d(x_t, \hat{x}_t)$. The triangle inequality, Lipschitz property, and registered one-step discrepancy yield

$$\begin{aligned}
 e_{t+1} &= d(F_t(x_t), \hat{F}_t(\hat{x}_t)) \\
 &\leq d(F_t(x_t), F_t(\hat{x}_t)) + d(F_t(\hat{x}_t), \hat{F}_t(\hat{x}_t)) \\
 &\leq L_t e_t + \kappa_t.
 \end{aligned}$$

Repeated substitution gives Equation (5). Here κ_t bounds the second distance at the candidate-side point $\hat{x}_t$ on the registered paired path in the same metric d . This deterministic derivation is withdrawn if either trajectory leaves the region on which L_t and κ_t are valid or if the local discrepancy was bounded on a different path.

A.3 Cross-scale composition

Set

$$\Gamma_{\ell:\ell} = \text{id}_{X_\ell}, \quad \Gamma_{\ell:\ell+q} = \Gamma_{\ell+q-1} \circ \cdots \circ \Gamma_\ell \quad (q \geq 1).$$

For every level used below, register the nonempty domain

$$A_j^{\text{comm}} = \{z \in A_j \cap B_j : F_j(z) \in B_j, \Gamma_j(z) \in A_{j+1}\},$$

so both compositions in the definition of η_j are typed. For the particular $x \in A_\ell^{\text{comm}}$, assume for every $q = 0, \dots, r-1$ that $\Gamma_{\ell:\ell+q}(x) \in A_{\ell+q}^{\text{comm}}$, and assume both $\Gamma_{\ell:\ell+q}F_\ell(x)$ and $F_{\ell+q}\Gamma_{\ell:\ell+q}(x)$ lie in the registered region on which $\Gamma_{\ell+q}$ has constant $G_{\ell+q}$. For $r=1$, Equation (6) is exactly the definition of η_ℓ . Assume the result for r .

Insert $\Gamma_{\ell+r}F_{\ell+r}\Gamma_{\ell:\ell+r}(x)$ between the two terms at level $\ell + r + 1$. The Lipschitz property of $\Gamma_{\ell+r}$ multiplies the existing r -level discrepancy by $G_{\ell+r}$, while the newly inserted one-step commutation term is at most $\eta_{\ell+r}$. Hence

$$E_{r+1} \leq G_{\ell+r}E_r + \eta_{\ell+r},$$

which expands to Equation (6) for $r + 1$. This induction proves the displayed weighted sum. The empty product preserves the newest defect with coefficient one. There is no initial mismatch because both sides begin at the same x ; different starting states invoke the separate path bound. Every composition must type-check, and every quantified intermediate image and paired point must satisfy its registered admissibility/Lipschitz premise. If even one premise is false or unrecorded, the only valid output is `CROSS_SCALE_BOUND_WITHDRAWN`; no finite bound may be returned.

A.4 Exact-binomial coverage and the zero-violation threshold

For fixed $C, F_C, D, \varepsilon_{\text{cert}}, Q$, let $W_i \stackrel{\text{iid}}{\sim} Q$ under the reference law and define

$$V_i = \mathbf{1}\{D(W_i) > \varepsilon_{\text{cert}}\}$$

so that the indicators V_i are iid Bernoulli with parameter $p_Q = \Pr_Q\{D(W) > \varepsilon_{\text{cert}}\}$. The one-sided Clopper–Pearson construction in Equation (7) inverts an exact binomial test and has coverage at least $1 - \beta$ [Clopper and Pearson, 1934]. Consequently, the decision $U_{\text{CP}} \leq \delta_{\text{cert}}$ supports $p_Q \leq \delta_{\text{cert}}$ at that confidence level. At $K = 0$, solving $1 - \beta^{1/m} \leq \delta_{\text{cert}}$ gives

$$m \geq \left\lceil \frac{\log \beta}{\log(1 - \delta_{\text{cert}})} \right\rceil.$$

Substitution of $\beta = \delta_{\text{cert}} = 0.05$ yields $m \geq 59$. This threshold is the minimum block count for the $K = 0$ design case; a design that permits $K > 0$ requires a separate calculation. It is a planning result under the iid reference law rather than an empirical estimate or an outcome-dependent stopping rule: the upper endpoint is 0.05033933833042015 at $m = 58$ and 0.049507609888227 at $m = 59$. The actual calibration and held-out counts were each fixed at 64; neither stopped after observing 59 zeroes. In the subsequently authorized first-and-only locked audit, A1, B0, B1, E, and F0 each observed zero adverse blocks among 64, giving the endpoint-wise one-sided 95% upper endpoint 0.04572970233076246 and descriptive two-sided 95% interval $[0, 0.05600908938663657]$. D2-CERT separately observed zero violations among 64, giving the endpoint-wise two-sided 95% interval $[0, 0.05600908938663657]$, a one-sided locked lower endpoint of zero, and no withdrawal. These intervals are not a family-wide 95% APHFS guarantee. For A1, B0, B1, E, and F0, the block event is a frozen constructed-workload implementation, accounting, fidelity, or decision-conformance event; the interval does not convert any of those endpoints into an estimate of open-world discovery, model-class power, search-policy superiority, or biological performance.

For D2-CERT, however, the registered scientific event is constructive. With the project’s Wolfram bit convention, Rule 170 returns the right-neighbor bit. Periodic boundaries therefore give $(F_{170}x)_j = x_{j+1 \bmod w}$, a cyclic permutation, so global density is exactly invariant for every binary initial state and time. The coarse predictor is identity and the event tolerance is zero. A violation is impossible for a correct implementation; the 0/64 calibration and 0/64 held-out records exercise certificate-state, role-separation, and paired simulator/fidelity machinery under this proved closure. Their CP calculations may describe a hypothetical block-level implementation/conformance-failure rate under the iid reference law, but not uncertainty about the density identity. This evidence class is *constructed closed-map certificate-pipeline conformance*; the immutable machine status remains `NOT_CONTRADICTED_BY_LOCKED_AUDIT`.

The actual successful role manifests follow this iid reference law conditioned on materializer acceptance. Calibration required 64 distinct 256-bit values; a union bound gives

$$q_{\text{cal}} \leq \binom{64}{2} 2^{-256} = 2016/2^{256} = 1.74105158070704 \times 10^{-74}.$$

The held-out role additionally had to avoid all 64 calibration values, giving

$$q_{\text{locked}} \leq \left[\binom{64}{2} + 64^2 \right] 2^{-256} = 6112/2^{256} = 5.27842622087372 \times 10^{-74}.$$

Conditioning means the implemented laws are not literally iid. By total variation, the reference-law coverage guarantees transfer conservatively as at least $1 - \beta - q_{\text{cal}}$ and $1 - \gamma - q_{\text{locked}}$. The tiny corrections alter no reported interval, threshold, decision, or status.

Calibration grants only the statement defined by Equation (7): for the frozen violation event under the iid reference law Q , the one-sided upper endpoint covers p_Q , and the certificate is granted only when that endpoint is at most δ_{cert} . The separately authorized locked audit evaluated a different question: whether a held-out, role-separated violation count contradicted $p_Q \leq \delta_{\text{cert}}$ under its frozen decision rule. It reports a two-sided exact interval and the one-sided lower endpoint in Equation (8). The certificate is withdrawn only when that lower endpoint exceeds δ_{cert} ; a raw locked fraction above δ_{cert} is not itself mathematical disproof. Failure to reject is labeled not contradicted, not confirmed.

Public-mock calibration reports only `PUBLIC MOCK CALIBRATION CRITERION MET` or its not-met counterpart and never grants a certificate. An actual grant is new-written as an immutable certificate record binding the approved freeze, runtime inventory, protocol/config/fidelity hashes, calibration role commitment, complete result bundle, separate β and γ , exact event, Q , and no-retuning declaration; the completed workflow created that record and completed its documented calibration-review step before locked authorization. Locked D2 is inapplicable without that granted record plus a schema-valid calibration-review approval bound to the raw result and certificate; it cannot grant a certificate. The four state fields `calibration_executed`, `locked_audit_executed`, `certificate_granted`, and `certificate_reviewed` are separate facts and may not be inferred from one another.

Post-hoc D2 withdrawal-rule operating characteristics. This paragraph and Table 3 carry the labels `POST_HOC_OPERATING_CHARACTERISTIC_ANALYSIS`, `NOT_A_PREREGISTERED_ENDPOINT`, and `DOES_NOT_CHANGE_D2_STATUS`. For $m_L = 64$, $\gamma = \delta_{\text{cert}} = 0.05$, the frozen one-sided lower-bound rule first satisfies $L_{\text{CP}}(K_L, 64; 0.95) > 0.05$ at $K_L = 7$:

$$L_{\text{CP}}(6, 64; 0.95) = 0.04161966005858663,$$

$$L_{\text{CP}}(7, 64; 0.95) = 0.05247175276586141.$$

Thus the trigger event is $K_L \geq 7$. Under a hypothetical true violation probability p , its withdrawal probability is the binomial tail

$$\Pr_p(K_L \geq 7) = 1 - \sum_{j=0}^6 \binom{64}{j} p^j (1-p)^{64-j}.$$

The domain is a hypothetical iid Bernoulli sequence of block-level implementation/event-mechanism failures governed by the unchanged rule. The calculation does not reuse role values, is not evidence about the observed blocks, and is not uncertainty about Rule 170’s analytical identity.

Table 3: Post-hoc operating characteristics of the unchanged D2-CERT withdrawal machinery under hypothetical block-level failures. Values are deterministic binomial-tail calculations, not new endpoint outcomes or evidence against the proved Rule 170 density identity.

Hypothetical block-failure rate p	Trigger	Withdrawal probability
0.05	$K_L \geq 7$	0.0402969
0.08	$K_L \geq 7$	0.2498734
0.10	$K_L \geq 7$	0.4609612
0.12	$K_L \geq 7$	0.6602532
0.15	$K_L \geq 7$	0.8635388
0.20	$K_L \geq 7$	0.9817801

Table 4: $K = 0$ sample-size planning at $\beta = 0.05$. Each entry is the smallest integer m satisfying $1 - \beta^{1/m} \leq \delta_{\text{cert}}$; designs allowing violations require a different calculation.

Target δ_{cert}	Minimum m for the $K = 0$ design
0.05	59
0.02	149
0.01	299
0.005	598
0.001	2995

Post-hoc multiplicity sensitivity. The registered endpoint-wise summaries remain controlling. As an illustrative `POST_HOC_MULTIPPLICITY_SENSITIVITY`, allocating a total one-sided tail probability 0.05 equally across the five homogeneous $K = 0, m = 64$ zero-adverse blockwise conformance endpoints under the iid reference law gives $\beta^* = 0.01$ and

$$1 - (0.01)^{1/64} = 0.06942795907030102.$$

This Bonferroni calculation requires each component interval’s own binomial assumptions but not independence among endpoints. It was not registered, changes no status, and is not substituted for the endpoint-wise 0.04572970233076246. A calculation treating all eleven endpoints as homogeneous is not reported as a family-wide result because five are deterministic or formula-conformance checks, D2-CERT uses a different direction and role, and the endpoints do not share one estimand.

A.5 Proposed APHFS-Q execution contract

This subsection is `PROPOSED_QUANTUM_EXTENSION` and `ROADMAP_ONLY`. It defines an interface for a future separately registered study; it reports no quantum execution, resource estimate, endpoint, or scientific result. The scientific logic—declared candidate language, complete accounting or disclosed sampling, operational ambiguity, class inadequacy, indeterminacy, fidelity withdrawal, and role-separated evaluation—is hardware-independent.

Table 5: Typed `QuantumResourceRecord` for a future execution contract.

Field	Registered content	Missing/failed action
semantics	Finite or sampled grammar, canonical encoding, valid-state rules, and unitary, CPTP channel, Hamiltonian, circuit, quantum-walk, or QCA update semantics.	Reject invalid candidates; do not impute a favorable status.
input_and_preparation	Input origin, normalization, preparation circuit/channel, construction cost, target error, success probability, repetitions, and reusability.	INDETERMINATE if correctness or cost is unsupported.
operator_access	Oracle, sparse access, LCU, block encoding, product formula, or other access model, including normalization, construction/query cost, error, and reuse assumptions.	Mark access-infeasible or INDETERMINATE.
logical_circuit	Logical qubits, fixed native logical gate basis, gate counts/depth, circuit family, synthesis tolerance, and termination budget. Toffoli is either native or compiled to T gates, never charged twice.	Mark resource-infeasible or reject an untyped record.

Field	Registered content	Missing/failed action
qec_mapping	Code family, distance, physical-noise model, decoder, factories, layout/connectivity, cycle time, and logical-failure allocation.	INDETERMINATE without an instance-specific mapping.
physical_estimate	Derived physical-qubit, cycle, wall-time, energy where available, and factory estimates, reported as ranges or confidence statements.	Do not present a point estimate as exact or mix it with logical counts.
measurement_and_readout	Observables, shots, estimators, simultaneous intervals, quantization grid, guard bands, device readout, and mitigation.	Return SIGNATURE_INDETERMINATE when an interval touches a cell boundary.
classical_postprocessing	Decoding, estimator evaluation, result verification, terminal-record materialization, storage, communication, and classical wall time.	Mark incomplete cost or output; do not hide classical work.
error_budget	Preparation, access, simulation, synthesis, logical/QEC, sampling, readout, classical-processing, and coarse-graining errors, each with domain and metric.	Withdraw the affected inference or return INDETERMINATE.
provenance	Candidate/config/software/hardware/calibration identifiers, compiler and mapping versions, hashes, failure records, and classical comparator.	No advantage, reproducibility, or domain claim without a complete binding.

Guard-banded signatures retain exact partition semantics only after statistical determination: each simultaneous interval must lie wholly inside one prespecified quantization cell; otherwise the candidate remains signature-indeterminate. This construction is a proposal, not a proved or tested theorem. Likewise, no generic formula such as independent, equal logical-gate failure is assumed. Probability terms may be combined by a valid proved bound or union bound only when their events and assumptions are registered; quantities in different metrics remain separate.

These typed records are compared componentwise, with Pareto dominance as the default. A scalar total is permitted only when its cost map, units, exchange rates, uncertainty treatment, and classical baseline were prespecified. Physical-qubit requirements are derived from **qec_mapping**, not stored as a union-type alternative to a QEC model. The native logical gate basis determines whether Toffoli is primitive or decomposed into T gates, preventing double counting. Preparation, access, measurement/readout, terminal-record output, verification, storage, and classical wall time are part of the end-to-end resource statement rather than free peripherals.

Candidate classes must also state which physical premises they adopt: locality/causality, no-signalling, symmetry/isotropy, unitarity or CPTP dynamics, conservation, continuum-limit dispersion, and the relevant scattering or spectroscopic constraints. Bell’s theorem constrains local hidden-variable accounts under its assumptions [Bell, 1964]; it does not force all pre-particle grammars to be QCA. QCA remains one candidate family with substantial direct prior art [Watrous, 1995, Meyer, 1996, Białynicki-Birula, 1994, Schumacher and Werner, 2004, Arrighi et al., 2011, Gross et al., 2012, D’Ariano and Perinotti, 2014, 2017, Farrelly, 2020].

Table 6: Falsifiable milestones for a future quantum execution program.

Stage	Candidate object	Required evidence	Failure action
Q0	Small local unitary or QCA	Exact classical small-instance encoding cross-check	Reject encoding
Q1	Fixed quantum candidate	Instance-specific logical qubits, depth, non-Clifford count, shots, and QEC model	Mark resource-infeasible
Q2	Registered observable	Classical/quantum comparison for the same observable and tolerance	Withdraw fidelity
Q3	Finite quantum-native grammar	New-role ambiguity and whole-class gate	Retain set, reject class, or abstain
Q4	Physical or chemical signature	Independently collected domain observations	Reject or revise the candidate class
Q5	Reaction or molecular class	Separate publicly preregistered future biological study	No biological claim from earlier stages

Table 7: Concrete direct-method comparison. This replaces broad capability checkmarks with the premise and conclusion of a cited procedure; no row assigns APHFS an intrinsic advantage.

Procedure	Direct precedent and premise	Relation to the proposed APHFS integration
Cellular-automaton coarse-graining	Israeli and Goldenfeld construct coarse descriptions and study predictability for cellular automata [Israeli and Goldenfeld, 2004, 2006].	Directly precedes coarse-graining of ECA. APHFS adds a proposed requirement to bind acceptance to stated domains, upper observables, tolerances, and decisions; the present ECA checks are not biological multiscale validation.
Version spaces and candidate elimination	Candidate elimination maintains the hypotheses consistent with examples, and generalization is framed as search in a hypothesis space [Mitchell, 1977, 1982].	Directly precedes retained candidate sets. APHFS’s additional proposal is address-level terminal accounting and explicit failure/class-inadequacy semantics, not priority for set-valued search.
Finite-program synthesis	Sketching and syntax-guided synthesis search grammar-constrained programs; unrealizability logic can prove that no program in a grammar satisfies a specification [Solar-Lezama et al., 2006, Alur et al., 2013, Kim et al., 2023].	These are formal-specification procedures. APHFS instead proposes separating an executable grammar from empirical evidence and statistical class bounds; synthesis realizability is not empirical adequacy.
Partial identification and model confidence sets	Partial identification and model-confidence-set procedures return sets under their own sampling and decision assumptions [Manski, 2003, Hansen et al., 2011].	They directly precede set-valued conclusions. APHFS operational signatures, retained candidates, and class-inadequacy gate have different estimands and must not be presented as replacements for those methods.
Goal-oriented error control	Quantity-of-interest error estimation, including a lattice-model extension, protects selected outputs rather than every microscopic value [Prudhomme and Oden, 1999, Oden et al., 2005].	This directly precedes observable-relative numerical control. APHFS proposes linking it to candidate decisions and failure records; it does not claim the underlying principle as new.

Procedure	Direct precedent and premise	Relation to the proposed APHFS integration
Systems-biology identifiability	Structural and practical identifiability analyze whether parameters or combinations are determined under a specified model and observation design [Bellman and Åström, 1970, Chis et al., 2011, Gutenkunst et al., 2007, Raue et al., 2009, Villaverde et al., 2016].	APHFS’s finite whole-candidate signature is a neighboring operational object, not a substitute for parameter identifiability and not biological validation.
QCA and discrete quantum dynamics	One-dimensional QCA, quantum lattice gases, localizability, and index theory formalize specific local quantum dynamics [Watrous, 1995, Meyer, 1996, Arrighi et al., 2011, Gross et al., 2012].	These are direct predecessors of any future quantum candidate grammar. APHFS claims only a proposed accounting, failure, fidelity, and evidence interface around such candidates.
End-to-end fault-tolerant resources	State preparation, surface-code mapping, and instance-level resource studies expose costs omitted by query counts [Plesch and Brukner, 2011, Fowler et al., 2012, Gidney and Ekerå, 2021, Low and Chuang, 2019, Reiher et al., 2017, Liu et al., 2024, Ek-lund et al., 2026].	They motivate the typed resource record. No APHFS quantum implementation, instance estimate, or advantage was performed here.

A.6 Frozen split roles

The completed protected workflow retained the frozen role separation. D2-CERT alone used 64 calibration blocks. A1, B0, B1, D2-CERT, E, and F0 each used 64 locked blocks; A0, C, D0, D1, and D2-MEM used their registered deterministic or complete finite censuses. The calibration role was materialized once and calibration was executed once in the earlier authorized phase. The locked role was materialized once and the locked audit was executed once. Neither role was rematerialized, replaced, or reordered; neither protected execution was retried. Raw role values remained sealed through confirmatory execution, separate result review, and independent read-only recomputation of safe artifacts. They were approved for subsequent release only as `POST_AUDIT_REPRODUCTION_ONLY`; any replay is non-confirmatory and cannot replace the immutable result. No such replay was run for this release candidate.

Role materialization enforced within-role uniqueness and calibration–locked nonintersection. Consequently, the implemented role laws are the ideal iid reference laws conditioned on no collision. The union-bound total-variation deviations are at most $q_{\text{cal}} \leq 1.74105158070704 \times 10^{-74}$ and $q_{\text{locked}} \leq 5.27842622087372 \times 10^{-74}$. These disclosed conditioning terms do not change any numerator, denominator, Clopper–Pearson interval, threshold, or decision.

Table 8: Completed data-role separation and prohibited reuse.

Role	Permitted use	Prohibited use
Development	Choose grammar features, probes, quantizers, maps, thresholds, and proposed filter order.	Claim locked performance or calibrate the final Clopper–Pearson certificate.

Role	Permitted use	Prohibited use
Calibration	For D2-CERT only, evaluate the already frozen violation event under registered Q ; compute U_{CP} .	Modify $C, F_C, D, \varepsilon_{cert}, Q$, probes, memory lag, or the certificate threshold.
Locked audit	Evaluate the complete frozen decision pipeline once under the signed manifest; report the exact interval and frozen withdrawal decision.	Select candidates, tune thresholds, repair failures, or rerun until favorable.
Exploratory follow-up	Diagnose a failure and propose a new version.	Replace the original locked result or inherit its confirmatory label.

A.7 Complete frozen endpoint and fidelity protocol

Table 9 preserves the implementation-facing details moved out of the main scientific narrative. Nothing in this registry was changed after the locked outcome.

For A0, *production/reference* denotes implementation-path separation, not independent origin. The scalar reference uses an explicit 111-to-000 rule-string lookup and a per-cell update; the NumPy-vectorized production path uses bit indexing. They do not share rule-lookup or one-step update functions, but both were developed within the same project under the same specification with substantive AI assistance. The comparison is therefore an internal deterministic implementation-conformance check, not an externally authored implementation or independent replication. Legacy frozen field names containing *independent* are retained as immutable schema labels and are interpreted only in this narrower sense. The frozen boundary token **reflect** implements edge replication: an out-of-range neighbor is set to the nearest boundary cell, not to the next interior cell of a mirror-reflected array. The scientific description therefore uses *edge-replicate* while retaining the historical token where exact code/configuration identity matters.

Table 9: Complete frozen benchmark protocol and failure interpretation.

ID	Registered target	Observation unit and sample	Decision rule	Failure interpretation
A0/A1	Complete 256-rule ECA accounting; data-generating-rule and signature-class retention	A0 is a deterministic 256-rule truth-table and paired production/reference trajectory census over periodic, fixed-zero, fixed-one, and edge-replicate boundaries (historical code token: reflect). A1 uses 64 locked role-derived blocks under its separately registered boundary/trajectory interface and no calibration blocks. Blocks, not cells or times, are the observation units. Its reference block law is iid; the implemented law is that law conditioned on role uniqueness and calibration nonoverlap.	Exact candidate losses and a constructed-retention conformance event; endpoint-wise one-sided 95% CP upper bound under the iid reference law.	Missing candidate/class, generating-class omission, failed execution, fidelity instability, or incomplete ledger is adverse. Multiplicity is retained rather than forced to one.
B0/B1	Avoid false class inadequacy in-class and detect the constructed $y = o_L \oplus o_R \oplus f_\theta(l, c, r)$ family	Each role value creates one distinct complete-support block; 64 locked blocks and no calibration blocks per endpoint. These are constructed workloads under the same reference/conditioned role-law distinction.	Exact complete-support candidate losses and singleton-class minima; B1 threshold 0.2; endpoint-wise CP conformance summary across blocks.	Missing coverage or an invalid bound is indeterminate. False B0 rejection or a retained/ indeterminate B1 decision is adverse.
C	Path-bound conformance for contractive, neutral, and expanding maps under three perturbation types	Nine deterministic constructed cases and 81 comparison points; no calibration or locked randomness.	Deterministic inequality with frozen 10^{-12} comparison tolerance; domain exit is never clipped.	Any bound violation or unregistered domain exit is adverse.

ID	Registered target	Observation unit and sample	Decision rule	Failure interpretation
D0–D2	Rule 204 identity, Rule 90 density non-closure, Rule 170 global-density certificate machinery, and two-bit memory repair	D0, D1, and D2-MEM are deterministic. D2-CERT alone uses 64 calibration and 64 locked blocks under the reference/conditioned role-law distinction.	D2-CERT uses Equations (7) and (8); D2-MEM requires lag-0 ambiguity and exact lag-1 repair.	D2-CERT is constructed closed-map certificate-pipeline conformance under the provably closed periodic Rule 170 density map; its exact status is a distribution-conditional non-withdrawal state, not confirmation. Missing support, fidelity instability, or incomplete records cannot pass.
E	Compare exhaustive, fixed-order, development-frozen-order, and adaptive-fidelity policies	The same truth-table observation stream and 64 locked blocks; no calibration blocks. Both static orders are distinct salted-SHA-256 pseudo-random permutations fixed before protected execution; neither encodes a scientific ordering prior.	Every candidate-block step, ordering action, probe, reference refinement, retry, and termination cost is billed under the registered bookkeeping map. Policy agreement is constructed-workload decision conformance; totals and policy ranking are descriptive and not wall-clock estimates.	Decision disagreement, unbilled access in the recorded trace, budget breach, missing trace, or unstable fidelity is adverse.
F0	Recover project-defined Rule 54/110 whole-trajectory fixtures	Each of 64 locked blocks draws one project-defined fixture case from the iid reference mixture; the implemented law is conditioned on role acceptance. Realized counts were 32 clean translations, 16 perturbations, and 16 negative controls.	Endpoint-wise constructed-fixture conformance-failure summary under the iid reference law; realized strata are descriptive.	No velocity, lifetime, held-out prediction, persistent detection, literature-seed provenance, or novel-discovery claim is registered.

Table 10: Benchmark-specific fidelity profiles. Sufficiency is assessed at protected upper-level observables and decisions, not by raw datatype proximity.

Profile	Registered/reference tiers	Protected observable	Failure action
A1 / D0–D2	Same-block vector/scalar ECA evaluators; fixed Rule 204/90/170 identities	Candidate/class retention, complete microstate, density event, or memory-lag decision	Preserve paired records; instability is adverse; missing refinement is indeterminate.
B0 / B1 / C	Bit-index vs explicit-string; vector vs scalar; decimal-6 float vs decimal-12 Decimal	Exact candidate losses, class decision, bound violation, and domain exit	Missing support or tier disagreement is indeterminate; domain exit withdraws; never clip.
E / F0	Lazy cheap/full/reference evaluation; vector/scalar template conformance	Policy decision, best candidate, complete cost ledger, and fixture/control decision	Hidden cost, tier disagreement, or changed decision is adverse.

A.8 Candidate, gate, and look ledger

The registered machine-readable candidate and look ledgers contain at least:

```
(run_id, config_hash, source_hash, candidate_id, description_bits, rule_id,
 parse_status, execution_status, signature_id, representative_id, resource_usage, gate_id,
 look, sample_count, split_id, alpha_allocated, bound_lower, bound_upper,
 threshold, decision, seed_block, precision_id, failure_code, timestamp).
```

This tuple is the general framework interface, not a claim that every field drove the immutable study. The repeated-look construction in Equation (9) was not in the protected execution call graph. In that equation the radius denominator is the cumulative sample size n_s available at look s , not the look index s itself. Protected B0/B1 instead used exact complete-support candidate losses, and every registered signature class in those gates was a singleton. The frozen `alpha_ledger_complete` field records complete 256-candidate coverage; it does not establish that the repeated-look allocation was exercised. Consequently, the class minima in Equation (4) equal their candidate losses here. Neither its non-singleton aggregation case nor Equation (9)’s repeated-look radius generated

an immutable endpoint result. The release-source correction is recorded as `POST_RESULT_FORMAL_CORRECTION` and `NO_LOCKED_ENDPOINT_DEPENDED_ON_THIS_DISPLAY`; the historical frozen source remains unchanged.

In an exhaustive run, each addressable description has exactly one terminal `execution_status` under the accounting rule in Section 3. The ECA registry has exactly 256 rule IDs. `CANONICAL_DUPLICATE` requires a representative pointer and proof of equality on the full registered evaluation interface; finite-probe signature membership alone never removes a candidate from the locked estimand.

Each fidelity-refinement ledger row uses the same coupled observation block and must additionally contain the lower-level precision or resolution change, the protected upper-level observable, its frozen error tolerance, the time horizon, the initial-state or environment distribution, the observable values before and after refinement, the resulting observable change, the registered decisions before and after refinement, whether that decision changed, and whether another refinement is required. Raw float32–float64 proximity without this observable- and decision-level check is not a fidelity certificate.

The canonical observable vocabulary uses `final_state_density`; the hyphenated or space-separated spelling is invalid. All config, protocol, role-manifest, authorization, and result-bundle boundaries are validated with JSON Schema Draft 2020-12 before use and after output.

Equation (9) covers only registered frozen candidates and gates. If the candidate universe is countable and adaptively revealed, each candidate must receive a nonnegative preassigned weight w_h with $\sum_h w_h \leq 1$, replacing $1/N$ by w_h , before its data are inspected. Otherwise a new error budget and a new confirmatory split are required.

The immutable result contains 718 block records and 452 fidelity records. All 452 fidelity rows are `PASS`; no row has a changed protected decision, an observable change beyond its registered tolerance, or a requirement for further refinement. These are observable- and decision-level fidelity checks, not a claim based only on raw float agreement.

A.9 First-and-only locked endpoint registry

Table 11 preserves all eleven registered endpoint statuses without collapsing their different evidentiary roles. Deterministic and formula-conformance controls do not receive pseudo-replicated Clopper–Pearson intervals. Every interval shown for a 64-block endpoint is endpoint-wise.

Table 11: First-and-only locked endpoint outcomes.

Endpoint	Evidence category	Event count	Exact rule	interval or frozen	Status
A0	Deterministic conformance	0/256	Complete CP not applicable	256-rule census;	PASS
A1	Constructed-workload retention conformance	0/64	One-sided = 0.04572970233076246; two-sided [0, 0.05600908938663657]	upper 95%	PASS
B0	Constructed in-class gate conformance	0/64	One-sided = 0.04572970233076246; two-sided [0, 0.05600908938663657]	upper 95%	PASS
B1	Constructed out-of-class gate conformance	0/64	One-sided = 0.04572970233076246; two-sided [0, 0.05600908938663657]	upper 95%	PASS

Endpoint	Evidence category	category	Event count	Exact interval or frozen rule	Status
C	Formula/domain conformance		0/9	Registered nine-case census; CP not applicable	PASS
D0	Deterministic identity		0/64	Rule 204 full-microstate identity; CP not applicable	PASS
D1	Explicit formula witness		0/1	Rule 90 non-closure witness; CP not applicable	PASS
D2-CERT	Constructed closed-map certificate-pipeline conformance		0/64	Two-sided 95% [0, 0.05600908938663657]; locked lower = 0; no withdrawal	NOT_CONTRADICTED_BY_LOCKED_AUDIT
D2-MEM	Complete finite-state formula census		0/4	Lag 0 ambiguous; lag 1 exact; CP not applicable	PASS
E	Constructed policy-decision conformance		0/64	One-sided upper = 0.04572970233076246; two-sided 95% [0, 0.05600908938663657]	PASS
F0	Constructed fixture conformance		0/64	One-sided upper = 0.04572970233076246; two-sided 95% [0, 0.05600908938663657]	PASS

For the five constructed-workload endpoints, the displayed exact-binomial intervals summarize the rate of their registered blockwise adverse conformance event under the iid reference law. They do not change the evidence category, and the implemented role law retains the conditioning correction stated above. D2-CERT’s interval analogously audits the certificate machinery under hypothetical block-level conformance failures; the underlying Rule 170 density identity is proved, not estimated from 64 blocks.

The result-specific interpretations are deliberately narrower than the status column:

1. **A / conformance and ambiguity.** A0 recorded zero mismatches in the complete 256-rule truth-table and four-boundary production/reference trajectory census and formed 36 registered canonical signature classes. A1 retained the data-generating rule and generating signature class in all 64 blocks, with zero adverse blocks. It did not force unique identification: 59 blocks retained one rule and five retained two; 61 blocks retained one signature class and three retained two. Cells and time points were not treated as iid replicates. The 36 classes are induced by a specific operational interface: width 32, eight updates, periodic trajectories from, in order, the single-center and alternating probes, with final density followed by whole-trajectory mean density for each probe. Quantizer boundaries 0.125, 0.25, 0.5, 0.75, 0.875 define half-open bins with exact boundary ties assigned to the upper-index bin; a missing value would use code `−32768`. The four signed 16-bit bin codes and canonical full specification header are serialized under `eca-signature-v1` as the magic prefix `APHFS-SIG\0`, a four-byte little-endian header length, the canonical JSON header, and then the four little-endian signed 16-bit codes; the resulting bytes are hashed with SHA-256. The other three A0 boundary conditions test simulator conformance but do not enter the signature. These are therefore not universal ECA dynamical-equivalence or standard symmetry classes. For a correct implementation, A1’s zero adverse count follows from noiseless generation by an enumerated rule and retention of every exact-zero-loss candidate; its block interval concerns implementation/accounting conformance, while the scientifically informative outcome is the residual

operational ambiguity.

2. **B / inadequacy.** B0 returned `RETAIN_CLASS` in all 64 in-class complete-support blocks, with class minimum lower bound zero and no false `MODEL_CLASS_INADEQUATE` decision. B1 returned `MODEL_CLASS_INADEQUATE` in all 64 complete radius-two blocks; every registered class minimum lower bound was 0.5, above the frozen threshold 0.2. B1 is a constructed distinguishable out-of-class family: its outer-XOR construction makes every radius-one candidate disagree on exactly half of the complete 32-row support. B0's truth candidate has exact loss zero. Because the registered classes in these gates are singletons, class minima equal exact candidate losses; neither a non-singleton aggregation nor the general repeated-look radius was exercised. The 64 role-derived repetitions are gate/ledger conformance checks, not evidence that APHFS detects arbitrary model misspecification.
3. **C / formula and domain conformance.** All nine registered cases had zero bound violations and zero domain exits. The largest raw absolute reference-candidate discrepancy was 0.16618718314740488 and is meaningful only relative to its registered bound. A smallest margin near -10^{-16} remained within the frozen 10^{-12} comparison tolerance and did not change a protected decision. This is a nine-case implementation audit, not a universal numerical-accuracy result.
4. **D / coarse dynamics.** D0 reproduced Rule 204 full-microstate identity. In the registered Rule 90 non-closure witness, microstates $[1, 1, 0, 0]$ and $[1, 0, 1, 0]$ both had current density 0.5, but their next densities were 1 and 0, respectively. D2-MEM completed the four-state two-bit census with coarse sequences 0000, 0101, 1010, and 1111; lag 0 was ambiguous and lag 1 was exact. The previously granted Rule 170 certificate, subsequently approved through the documented calibration-review step, entered a D2-CERT machinery-conformance audit on a provably closed map. Under periodic boundaries Rule 170 is a cyclic shift, so global density is preserved for every binary state and time. The certificate was not contradicted by the D2-CERT locked audit: zero violations among 64 yielded locked lower endpoint zero and did not trigger withdrawal. The exact machine status is `NOT_CONTRADICTED_BY_LOCKED_AUDIT`. This exercises the certificate state machine, paired simulators, fidelity checks, and role separation; it does not use 64 blocks to estimate whether the density identity is true, and the locked result did not grant, prove, or universally validate a new certificate.
5. **E / policy agreement and cost.** All four policies matched exhaustive search on the final decision, best valid candidate, and failure status in all 64 blocks. The frozen meter recorded no unbilled loss access or budget overrun. Both `fixed_order` and `development_frozen_order` are distinct permutations of the same 256 candidates, obtained before protected execution by sorting differently salted SHA-256 digests; neither is a scientific prior or a learned ranking. All policies used the same per-block complete truth-table observation stream. The no-unbilled-access statement is bounded to the frozen instrumented code path and retained trace; it does not claim to rule out a hypothetical external cache outside that interface. Total registered cost units were 132,096 for exhaustive, 75,080 for `fixed_order`, 89,816 for `development_frozen_order`, and 92,592 for adaptive fidelity. Relative to exhaustive, these were 43.16%, 32.01%, and 29.91% lower for fixed order, development-frozen order, and adaptive fidelity, respectively. The immutable policy labeled `fixed_order` was cheapest on this constructed workload and registered bookkeeping map. Adaptive fidelity did not win this finite grammar and carried about 23.32% more cost than fixed order and about 3.09% more than the development-frozen order. The units are not wall-clock time, and this comparison supports neither a universal ordering policy nor a rejection of multifidelity methods.
6. **F0 / known-template conformance.** The 64 project-defined Rule 54/110 fixture blocks comprised 32 clean translations, 16 single-cell perturbations, and 16 negative controls. These realized strata are descriptive. Clean similarity was 1; perturbation similarity was 0.9987980769230769; and negative-control similarity ranged from 0.5108173076923077 to 0.5540865384615384. Positive cases replay project-defined registered whole-trajectory templates and negative controls use a project-defined digest construction. F0 therefore remains constructed known-fixture conformance, not velocity, lifetime, held-out prediction, persistent-structure detection, literature-seed reproduction, or novel discovery.

Table 12: Exact Benchmark E registered bookkeeping-cost composition. The five billed components sum to each immutable total; retry/other was zero for every policy. The two static orders are distinct prespecified salted-SHA-256 pseudo-random permutations, and the totals are not wall-clock measurements.

Policy	Candidate	Ordering	Probes	Reference	Retry/other	Total
<code>fixed_order</code>	57,672	16,384	512	512	0	75,080
<code>development_frozen_order</code>	72,408	16,384	512	512	0	89,816
Adaptive fidelity	70,360	16,384	512	5,336	0	92,592
Exhaustive	131,072	0	512	512	0	132,096

Table 13: A1 retained ambiguity. Every block retained the registered data-generating rule; multiplicity records honest non-identifiability under the finite observation interface.

Retained object	Multiplicity	Number of blocks
Candidate	1	59
Candidate	2	5
Signature class	1	61
Signature class	2	3

Table 14: Descriptive F0 known-template strata. These are not separate inferential endpoints.

Stratum	Blocks	Similarity
Registered translation	32	1
Single-cell perturbation	16	0.9987980769230769
Negative control	16	0.5108173076923077–0.5540865384615384

A.10 One-time provenance and empty failure ledger

The unique formal command was `.venv-protected-freeze-v1/bin/python -I scripts/run_locked_audit_once.py`. It was invoked once, exited with code zero, and recorded a process duration of 12.904217 seconds. The retry flag was false and the permanent guard was retained. Independent read-only recomputation of released safe artifacts reviewed all 718 block records and 452 fidelity records and found zero discrepancy. The scientific engine was not invoked during that recomputation.

Table 15: Safe commitment chain for the first-and-only locked execution. Hashes bind artifacts; they do not disclose role values.

Artifact	SHA-256
Approved final freeze	ff4313397621301978fe296845b4aebecd30caae771bc880ebfae4d5b6939b4c
Cumulative non-scientific amendment	af59253b4f8ac9e9c570120712462daed52322b4aab0a57c8a61b7a39cf5b949
Amendment-review approval	90d46ec748908d919716190e507b2c1cb3ca2c888900121952954b9e0ba5a450
Amended runtime inventory	6ac8077cac3fa61ce05380ec93531fd63b63525ca83045815e9bc69fd77862b1
Protected runtime environment	4bf6a93516fe3af6c5df828fd1b140812d178f94bb84891e700e015025ed7ed9
Protocol	5c28bab547056fb188e5e24c9ff3f26aefc5b85117aa01439e362cf7f72ad8527
Configuration	23da043818dd11ac5a47dfb6a94198af333aa3673d543381f5fcb9b62a41ed9
Fidelity contract	24c28988441d9f53beb7976b60e3807ea27e3f38104a99d51af09887a6d9f137
Calibration result	71bad876e585867a76aa7ab07df4faa60d196220ef4e94e252c13fb6e93e0ad5
Calibration certificate	d39f19cc09b091cc41a63123ae8a72055bbe45d354cb0923b2374fd276f30fab
Calibration-review approval	c9bc66388d49de797a26ade4fc15892bf3518d7d1db37f1c59776b728d9a7d7a
Locked role commitment	c018abe17e6b9f5cbf80997bc03c086746867e1e87a58f0fea90f311a1782596
Locked-materialization approval	d549df48c99b711ab5146b930a12aa347804fe4636b3a2e89e68d6404a824325
Locked-execution approval	d2a1b18ace63bf83e22e33b171e9c7aa96ac470801cf7c266a7e1f3918ab7303
Locked provenance context	8a3299e2b89fe37c125209c6d43d48b624300807c630755d15db6f2244259050
Exact command commitment	cc09888f1e978ee2836032123278d34d5d00c702707e3dfb99bb8902d57ab97e
Execution intent	08a09c51de5d1e82dfb026a5b41126ba2373ddd30c6b9d38dcf9f0d372a9ab8b
Permanent guard	f16af7237fbb7dd46225864166ba30eb14eb5c2004eddf32573191705cfc4c85
Locked result	6fb3f08e48b4d3496e190fbc38b029ee7c15e327c504924f8c03b6e2083aec9c
Execution receipt	730bfe099a74b1ccfa281e653e9330bf91a5e73c30b24a58cf52dd88028c4766

These hashes bind the disclosed artifact contents and make later alteration detectable. The original freeze was not anchored to an external trusted timestamp or a public registration service before execution. The chain therefore supports content integrity and internal provenance, not independently verifiable third-party chronology. A future protected study should place its protocol and commitments under an external timestamp or public preregistration before any protected execution.

The immutable failure/indeterminate ledger has the registered columns shown in Table 16 and zero data rows. Empty means that no such record occurred in the immutable result; it does not mean that a failed row was deleted, imputed, or retried.

Table 16: Complete locked failure/indeterminate ledger.

Endpoint	Audit row ID	Source	Failure code	Counts as adverse
No rows in the immutable first-and-only locked result.				

The safe aggregates establish only conformance and registered machinery behavior in the complete finite 256-rule ECA grammar. They do not establish physical-law discovery, particles, molecules, cells, aging mechanisms, rejuvenation interventions, therapeutic benefit, quantum advantage, or component-level novelty. The biochemical and biological path remains a conditional roadmap.

A.11 Falsification and withdrawal matrix

Table 17: Statement-specific withdrawal conditions.

Statement	Withdrawal trigger	Required report
Finite-class oracle bound	Class, loss, or budget adapted on evaluation data; dependence or unbounded loss contradicts A1–A3	Mark theorem inapplicable; do not patch with a fixed-class formula.
Signature partition	Pairwise tolerance replaces exact serialized equality; quantizer changes mid-run; finite-probe equality is used for unsafe representative-only evaluation	Version a new partition, recompute every class, and restore candidate-level coverage.
Class inadequacy	Any candidate lacks a valid simultaneous lower bound; alpha ledger incomplete; deduplication changes the estimand	Return indeterminate, not <code>MODEL_CLASS_INADEQUATE</code> .
Path/cross-scale bound	Path leaves registered region; maps fail to type-check; a required coarse-map Lipschitz constant or commutation-defect domain is missing	Return <code>CROSS_SCALE_BOUND_WITHDRAWN</code> , report the failed premise and affected cases, and do not emit a finite cross-scale bound.
Coarse certificate	Calibration reused for tuning; Q shifts without bound; event changes; locked lower exact bound exceeds δ_{cert}	Withdraw certificate and recalibrate only after new registration and authorization.
Anytime elimination	Unregistered repeated looks or adaptively added candidates lack weights	Void affected eliminations and report alpha-budget breach.
Practical value	Immutable locked artifacts or safe source data fail independent read-only recomputation from released safe artifacts, or a later external-domain study fails	Preserve the original result and restrict the affected claim to the reproducible formal or finite-grammar evidence.

References

- Rajeev Alur, Rastislav Bodik, Garvit Juniwal, Milo M. K. Martin, Mukund Raghothaman, Sanjit A. Seshia, Rishabh Singh, Armando Solar-Lezama, Emina Torlak, and Abhishek Udupa. Syntax-guided synthesis. In *2013 Formal Methods in Computer-Aided Design*, pages 1–8. IEEE, 2013. doi: 10.1109/FMCAD.2013.6679385. URL <https://doi.org/10.1109/FMCAD.2013.6679385>.
- Pablo Arrighi. An overview of quantum cellular automata. *Natural Computing*, 18(4):885–899, 2019. doi: 10.1007/s11047-019-09762-6.
- Pablo Arrighi, Vincent Nesme, and Reinhard Werner. Unitarity plus causality implies localizability. *Journal of Computer and System Sciences*, 77(2):372–378, 2011. doi: 10.1016/j.jcss.2010.05.004. URL <https://doi.org/10.1016/j.jcss.2010.05.004>.
- Shivnath Babu, Rajeev Motwani, Kamesh Munagala, Itaru Nishizawa, and Jennifer Widom. Adaptive ordering of pipelined stream filters. In *Proceedings of SIGMOD 2004*, pages 407–418, 2004. doi: 10.1145/1007568.1007615.

- Andrew R. Barron and Thomas M. Cover. Minimum complexity density estimation. *IEEE Transactions on Information Theory*, 37(4):1034–1054, 1991. doi: 10.1109/18.86996.
- John S. Bell. On the einstein podolsky rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964. doi: 10.1103/PhysicsPhysiqueFizika.1.195.
- Richard Bellman and Karl J. Åström. On structural identifiability. *Mathematical Biosciences*, 7(3–4):329–339, 1970. doi: 10.1016/0025-5564(70)90132-X. URL [https://doi.org/10.1016/0025-5564\(70\)90132-X](https://doi.org/10.1016/0025-5564(70)90132-X).
- Charles H. Bennett, Ethan Bernstein, Gilles Brassard, and Umesh Vazirani. Strengths and weaknesses of quantum computing. *SIAM Journal on Computing*, 26(5):1510–1523, 1997. doi: 10.1137/S0097539796300933.
- Iwo Białynicki-Birula. Weyl, dirac, and maxwell equations on a lattice as unitary cellular automata. *Physical Review D*, 49(12):6920–6927, 1994. doi: 10.1103/PhysRevD.49.6920.
- Alessandro Bisio, Giacomo Mauro D’Ariano, and Alessandro Tosini. Quantum field as a quantum cellular automaton: The dirac free evolution in one dimension. *Annals of Physics*, 354:244–264, 2015. doi: 10.1016/j.aop.2014.12.016.
- Alessandro Bisio, Giacomo Mauro D’Ariano, and Paolo Perinotti. Quantum cellular automaton theory of light. *Annals of Physics*, 368:177–190, 2016. doi: 10.1016/j.aop.2016.02.009.
- Josh Bongard and Hod Lipson. Automated reverse engineering of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 104(24):9943–9948, 2007. doi: 10.1073/pnas.0609476104.
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016. doi: 10.1073/pnas.1517384113.
- Oana-Teodora Chis, Julio R. Banga, and Eva Balsa-Canto. Structural identifiability of systems biology models: A critical comparison of methods. *PLoS ONE*, 6(11):e27755, 2011. doi: 10.1371/journal.pone.0027755. URL <https://doi.org/10.1371/journal.pone.0027755>.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
- Matthew Cook. Universality in elementary cellular automata. *Complex Systems*, 15(1):1–40, 2004. doi: 10.25088/ComplexSystems.15.1.1.
- James P. Crutchfield and Karl Young. Inferring statistical complexity. *Physical Review Letters*, 63(2):105–108, 1989. doi: 10.1103/PhysRevLett.63.105.
- Giacomo Mauro D’Ariano and Paolo Perinotti. Derivation of the dirac equation from principles of information processing. *Physical Review A*, 90(6):062106, 2014. doi: 10.1103/PhysRevA.90.062106.
- Giacomo Mauro D’Ariano and Paolo Perinotti. Quantum cellular automata and free quantum field theory. *Frontiers of Physics*, 12(1):120301, 2017. doi: 10.1007/s11467-016-0616-z.
- David Deutsch. Quantum theory, the church–turing principle and the universal quantum computer. *Proceedings of the Royal Society A*, 400(1818):97–117, 1985. doi: 10.1098/rspa.1985.0070.
- Elliot C. Eklund, Arkin Tikku, Patrick Sinnott, William J. Huggins, Guang Hao Low, Dominic W. Berry, and Ivan Kassal. End-to-end simulation of chemical dynamics on a quantum computer, 2026. URL <https://arxiv.org/abs/2603.19007v1>. Version 1, submitted 19 March 2026; preprint, no journal reference listed as of 6 August 2026.
- Kevin Ellis, Catherine Wong, Maxwell Nye, Mathias Sablé-Meyer, Lucas Morales, Luke Hewitt, Luc Cary, Armando Solar-Lezama, and Joshua B. Tenenbaum. Dreamcoder: Growing generalizable, interpretable knowledge with wake–sleep bayesian program learning. In *Proceedings of PLDI 2021*, pages 835–850, 2021. doi: 10.1145/3453483.3454080.

- Terry Farrelly. A review of quantum cellular automata. *Quantum*, 4:368, 2020. doi: 10.22331/q-2020-11-30-368.
- Richard P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6-7): 467–488, 1982. doi: 10.1007/BF02650179.
- Austin G. Fowler, Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86(3):032324, 2012. doi: 10.1103/PhysRevA.86.032324. URL <https://doi.org/10.1103/PhysRevA.86.032324>.
- Edward Fredkin. Digital mechanics. *Physica D: Nonlinear Phenomena*, 45:254–270, 1990. doi: 10.1016/0167-2789(90)90186-S.
- Andrew Gelman, Xiao-Li Meng, and Hal Stern. Posterior predictive assessment of model fitness via realized discrepancies. *Statistica Sinica*, 6(4):733–807, 1996.
- Stuart Geman and Chii-Ruey Hwang. Nonparametric maximum likelihood estimation by the method of sieves. *The Annals of Statistics*, 10(2):401–414, 1982. doi: 10.1214/aos/1176345782.
- Craig Gidney and Martin Ekerå. How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits. *Quantum*, 5:433, 2021. doi: 10.22331/q-2021-04-15-433. URL <https://quantum-journal.org/papers/q-2021-04-15-433/>.
- Ulf Grenander. *Abstract Inference*. Wiley, 1981. ISBN 978-0-471-08267-5.
- David Gross, Vincent Nesme, Holger Vogts, and Reinhard F. Werner. Index theory of one dimensional quantum walks and cellular automata. *Communications in Mathematical Physics*, 310(2):419–454, 2012. doi: 10.1007/s00220-012-1423-1. URL <https://doi.org/10.1007/s00220-012-1423-1>.
- Lov K. Grover. A fast quantum mechanical algorithm for database search. In *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing*, pages 212–219, 1996. doi: 10.1145/237814.237866.
- Ryan N. Gutenkunst, Joshua J. Waterfall, Fergal P. Casey, Kevin S. Brown, Christopher R. Myers, and James P. Sethna. Universally sloppy parameter sensitivities in systems biology models. *PLoS Computational Biology*, 3(10):e189, 2007. doi: 10.1371/journal.pcbi.0030189.
- Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011. doi: 10.3982/ECTA5771. URL <https://doi.org/10.3982/ECTA5771>.
- Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
- Erik P. Hoel. When the map is better than the territory. *Entropy*, 19(5):188, 2017. doi: 10.3390/e19050188.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991.
- Marcus Hutter. *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, 2005. doi: 10.1007/b138233.
- Navot Israeli and Nigel Goldenfeld. Computational irreducibility and the predictability of complex physical systems. *Physical Review Letters*, 92(7):074105, 2004. doi: 10.1103/PhysRevLett.92.074105. URL <https://doi.org/10.1103/PhysRevLett.92.074105>.
- Navot Israeli and Nigel Goldenfeld. Coarse-graining of cellular automata, emergence, and the predictability of complex systems. *Physical Review E*, 73(2):026203, 2006. doi: 10.1103/PhysRevE.73.026203. URL <https://doi.org/10.1103/PhysRevE.73.026203>.
- Garud N. Iyengar. Robust dynamic programming. *Mathematics of Operations Research*, 30(2):257–280, 2005. doi: 10.1287/moor.1040.0129.

- Kevin Jamieson and Ameet Talwalkar. Non-stochastic best arm identification and hyperparameter optimization. *Proceedings of Machine Learning Research*, 51:240–248, 2016.
- Kirthevasan Kandasamy, Gautam Dasarathy, Jeff Schneider, and Barnabás Póczos. Multi-fidelity bayesian optimisation with continuous approximations. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1799–1808, 2017.
- Jonathan R. Karr, Jayodita C. Sanghvi, Derek N. Macklin, Miriam V. Gutschow, Jared M. Jacobs, Benjamin Bolival, Jr., Nacyra Assad-Garcia, John I. Glass, and Markus W. Covert. A whole-cell computational model predicts phenotype from genotype. *Cell*, 150(2):389–401, 2012. doi: 10.1016/j.cell.2012.05.044.
- Jinwoo Kim, Loris D’Antoni, and Thomas Reps. Unrealizability logic. *Proceedings of the ACM on Programming Languages*, 7(POPL):659–688, 2023. doi: 10.1145/3571216. URL <https://doi.org/10.1145/3571216>.
- Lisha Li, Kevin Jamieson, Giulia DeSalvo, Afshin Rostamizadeh, and Ameet Talwalkar. Hyperband: A novel bandit-based approach to hyperparameter optimization. *Journal of Machine Learning Research*, 18(185):1–52, 2018.
- Michael L. Littman, Richard S. Sutton, and Satinder Singh. Predictive representations of state. In *Advances in Neural Information Processing Systems 14*, pages 1555–1561, 2001. URL https://proceedings.neurips.cc/paper_files/paper/2001/hash/1e4d36177d71bbb3558e43af9577d70e-Abstract.html.
- Zhenning Liu, Xiantao Li, Chunhao Wang, and Jin-Peng Liu. Toward end-to-end quantum simulation for protein dynamics, 2024. URL <https://arxiv.org/abs/2411.03972v2>. Version 2, revised 16 April 2025; preprint, no journal reference listed as of 6 August 2026.
- Seth Lloyd. Universal quantum simulators. *Science*, 273(5278):1073–1078, 1996. doi: 10.1126/science.273.5278.1073.
- Carlos López-Otín, Maria A. Blasco, Linda Partridge, Manuel Serrano, and Guido Kroemer. The hallmarks of aging. *Cell*, 153(6):1194–1217, 2013. doi: 10.1016/j.cell.2013.05.039.
- Carlos López-Otín, Maria A. Blasco, Linda Partridge, Manuel Serrano, and Guido Kroemer. Hallmarks of aging: An expanding universe. *Cell*, 186(2):243–278, 2023. doi: 10.1016/j.cell.2022.11.001.
- Guang Hao Low and Isaac L. Chuang. Hamiltonian simulation by qubitization. *Quantum*, 3:163, 2019. doi: 10.22331/q-2019-07-12-163.
- Charles F. Manski. *Partial Identification of Probability Distributions*. Springer Series in Statistics. Springer New York, 2003. doi: 10.1007/b97478. URL <https://doi.org/10.1007/b97478>.
- Genaro J. Martínez, Andrew Adamatzky, and Harold V. McIntosh. Phenomenology of glider collisions in cellular automaton rule 54 and associated logical gates. *Chaos, Solitons & Fractals*, 28(1):100–111, 2006. doi: 10.1016/j.chaos.2005.05.013.
- David A. Meyer. From quantum cellular automata to quantum lattice gases. *Journal of Statistical Physics*, 85(5–6):551–574, 1996. doi: 10.1007/BF02199356. URL <https://doi.org/10.1007/BF02199356>.
- Tom M. Mitchell. Version spaces: A candidate elimination approach to rule learning. In *Proceedings of the Fifth International Joint Conference on Artificial Intelligence*, pages 305–310. International Joint Committee on Artificial Intelligence, 1977. URL <https://www.ijcai.org/Proceedings/77-1/Papers/048.pdf>.
- Tom M. Mitchell. Generalization as search. *Artificial Intelligence*, 18(2):203–226, 1982. doi: 10.1016/0004-3702(82)90040-6. URL [https://doi.org/10.1016/0004-3702\(82\)90040-6](https://doi.org/10.1016/0004-3702(82)90040-6).
- Hazime Mori. Transport, collective motion, and brownian motion. *Progress of Theoretical Physics*, 33(3):423–455, 1965. doi: 10.1143/PTP.33.423.

- Arnab Nilim and Laurent El Ghaoui. Robust control of markov decision processes with uncertain transition matrices. *Operations Research*, 53(5):780–798, 2005. doi: 10.1287/opre.1050.0216.
- J. Tinsley Oden, Serge Prudhomme, and Paul T. Bauman. On the extension of goal-oriented error estimation and hierarchical modeling to discrete lattice models. *Computer Methods in Applied Mechanics and Engineering*, 194(34–35):3668–3688, 2005. doi: 10.1016/j.cma.2004.08.010. URL <https://doi.org/10.1016/j.cma.2004.08.010>.
- Martin Plesch and Ćaslav Brukner. Quantum-state preparation with universal gate decompositions. *Physical Review A*, 83(3):032302, 2011. doi: 10.1103/PhysRevA.83.032302. URL <https://doi.org/10.1103/PhysRevA.83.032302>.
- Jan-Hendrik Prinz, Hao Wu, Marco Sarich, Bettina Keller, Martin Senne, Martin Held, John D. Chodera, Christof Schütte, and Frank Noé. Markov models of molecular kinetics: Generation and validation. *The Journal of Chemical Physics*, 134:174105, 2011. doi: 10.1063/1.3565032.
- Serge Prudhomme and J. Tinsley Oden. On goal-oriented error estimation for elliptic problems: Application to the control of pointwise errors. *Computer Methods in Applied Mechanics and Engineering*, 176(1–4):313–331, 1999. doi: 10.1016/S0045-7825(98)00343-0. URL [https://doi.org/10.1016/S0045-7825\(98\)00343-0](https://doi.org/10.1016/S0045-7825(98)00343-0).
- Andreas Raue, Clemens Kreutz, Thomas Maiwald, Julie Bachmann, Marcel Schilling, Ursula Klingmüller, and Jens Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, 2009. doi: 10.1093/bioinformatics/btp358.
- Markus Reiher, Nathan Wiebe, Krysta M. Svore, Dave Wecker, and Matthias Troyer. Elucidating reaction mechanisms on quantum computers. *Proceedings of the National Academy of Sciences*, 114(29):7555–7560, 2017. doi: 10.1073/pnas.1619152114.
- Fernando E. Rosas, Pedro A. M. Mediano, Henrik J. Jensen, Anil K. Seth, Adam B. Barrett, Robin L. Carhart-Harris, and Daniel Bor. Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data. *PLOS Computational Biology*, 16(12):e1008289, 2020. doi: 10.1371/journal.pcbi.1008289.
- Walter J. Scheirer, Anderson de Rezende Rocha, Archana Sapkota, and Terrance E. Boult. Toward open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(7):1757–1772, 2013. doi: 10.1109/TPAMI.2012.256.
- Jürgen Schmidhuber. Algorithmic theories of everything, 2000. Submitted 2000; revised 2002.
- Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324(5923):81–85, 2009. doi: 10.1126/science.1165893.
- Benjamin Schumacher and Reinhard F. Werner. Reversible quantum cellular automata, 2004. URL <https://arxiv.org/abs/quant-ph/0405174>. Preprint.
- Armando Solar-Lezama, Liviu Tancau, Rastislav Bodik, Vijay Saraswat, and Sanjit A. Seshia. Combinatorial sketching for finite programs. In *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems*, pages 404–415. Association for Computing Machinery, 2006. doi: 10.1145/1168857.1168907. URL <https://doi.org/10.1145/1168857.1168907>.
- Gerard ’t Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*, volume 185 of *Fundamental Theories of Physics*. Springer, 2016. doi: 10.1007/978-3-319-41285-6.
- Zane R. Thornburg, Andrew Maytin, Jiwoong Kwon, Troy A. Brier, Benjamin R. Gilbert, Enguang Fu, Yang-Le Gao, Jordan Quenneville, Tianyu Wu, Henry Li, Talia Long, Weria Pezeshkian, Lijie Sun, Daniela Matias de C. Bittencourt, John I. Glass, Angad P. Mehta, Taekjip Ha, and Zaida Luthey-Schulten. Bringing the genetically minimal cell to life on a computer in 4d. *Cell*, 189(9):2582–2597.e27, 2026a. doi: 10.1016/j.cell.2026.02.009.

- Zane R. Thornburg, Andrew Maytin, Jiwoong Kwon, Troy A. Brier, Benjamin R. Gilbert, Enguang Fu, Yang-Le Gao, Jordan Quenneville, Tianyu Wu, Henry Li, Talia Long, Weria Pezeshkian, Lijie Sun, Daniela Matias de C. Bittencourt, John I. Glass, Angad P. Mehta, Taekjip Ha, and Zaida Luthey-Schulten. Bringing the genetically minimal cell to life on a computer in 4d. *Cell*, 189(9):2780–2781, 2026b. doi: 10.1016/j.cell.2026.04.003. Published correction to doi:10.1016/j.cell.2026.02.009.
- Zane R. Thornburg et al. Fundamental behaviors emerge from simulations of a living minimal cell. *Cell*, 185(2): 345–360.e28, 2022. doi: 10.1016/j.cell.2021.12.025.
- Silviu-Marian Udrescu and Max Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020. doi: 10.1126/sciadv.aay2631.
- Vladimir N. Vapnik. *The Nature of Statistical Learning Theory*. Springer, 1995. doi: 10.1007/978-1-4757-2440-0.
- Alejandro F. Villaverde, Antonio Barreiro, and Antonis Papachristodoulou. Structural identifiability of dynamic systems biology models. *PLoS Computational Biology*, 12(10):e1005153, 2016. doi: 10.1371/journal.pcbi.1005153.
- John Watrous. On one-dimensional quantum cellular automata. In *Proceedings of IEEE 36th Annual Foundations of Computer Science*, pages 528–537. IEEE Computer Society Press, 1995. doi: 10.1109/SFCS.1995.492583. URL <https://doi.org/10.1109/SFCS.1995.492583>.
- Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society: Series B*, 86(1):1–27, 2024. doi: 10.1093/jrssl/bqkad009.
- Halbert White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982. doi: 10.2307/1912526.
- Stephen Wolfram. *A New Kind of Science*. Wolfram Media, 2002. ISBN 978-1-57955-008-0.
- Stephen Wolfram. A class of models with the potential to represent fundamental physics. *Complex Systems*, 29(2):107–536, 2020. doi: 10.25088/ComplexSystems.29.2.107.
- Yuling Yao, Aki Vehtari, Daniel Simpson, and Andrew Gelman. Using stacking to average bayesian predictive distributions. *Bayesian Analysis*, 13(3):917–1007, 2018. doi: 10.1214/17-BA1091.
- Konrad Zuse. *Rechnender Raum*. Vieweg, 1969. doi: 10.1007/978-3-663-02723-2. German edition; later translation metadata should not be conflated with this record.
- Robert Zwanzig. Memory effects in irreversible thermodynamics. *Physical Review*, 124(4):983–992, 1961. doi: 10.1103/PhysRev.124.983.
- Robert Zwanzig, K. S. J. Nordholm, and W. C. Mitchell. Memory effects in irreversible thermodynamics: Corrected derivation of transport equations. *Physical Review A*, 5(6):2680–2682, 1972. doi: 10.1103/PhysRevA.5.2680.